

人工智能安全

对抗攻击与防御

刁云峰

`diaoyunfeng@hfut.edu.cn`

目录

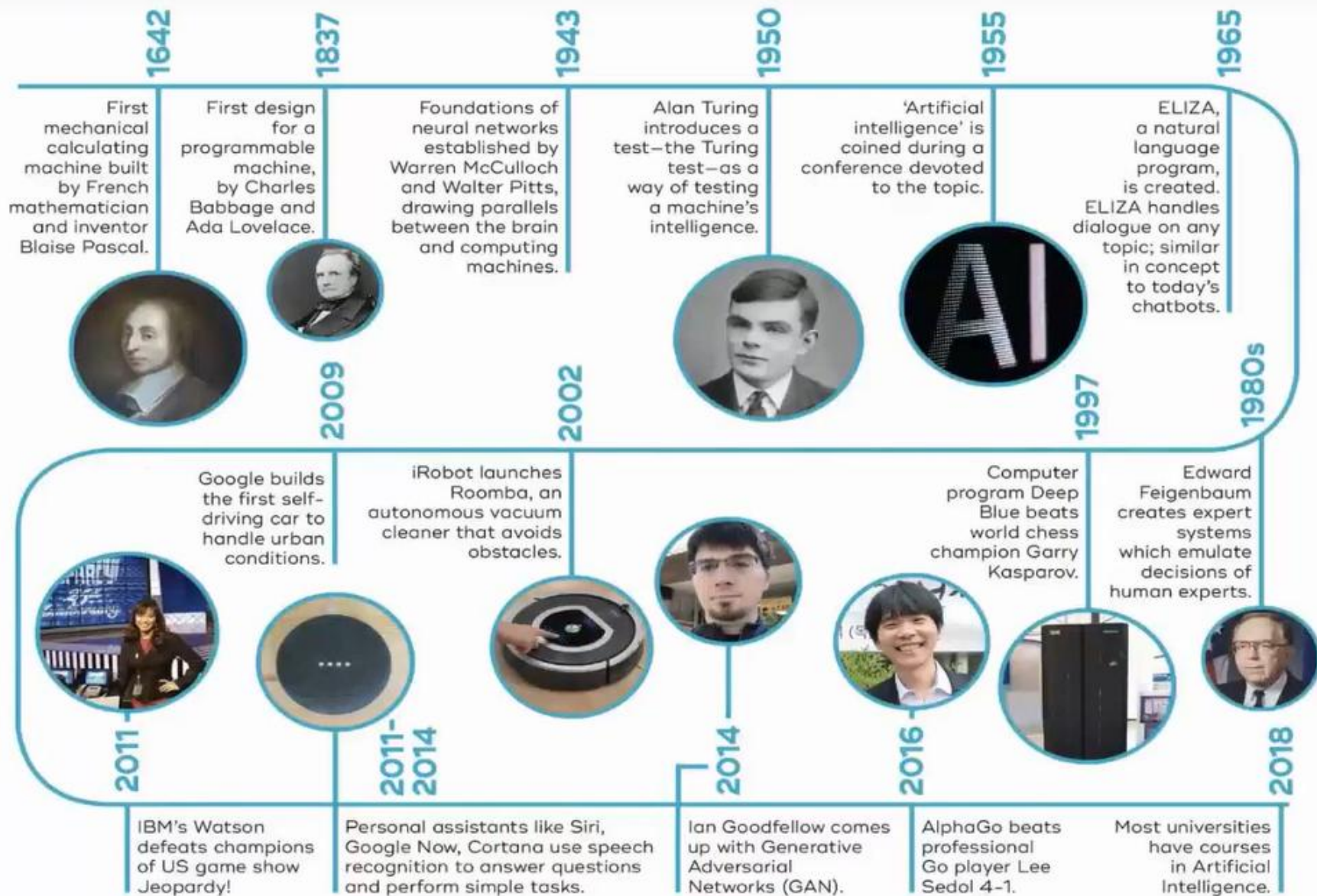
C O N T E N T S

- 01 | Background (背景)
AI对抗攻防背景介绍
- 02 | Literatures (相关实例)
AI对抗攻防的相关实例
- 03 | Methods (攻防原理)
AI对抗攻防部分算法原理

01 | Background (背景)

AI对抗攻防背景介绍

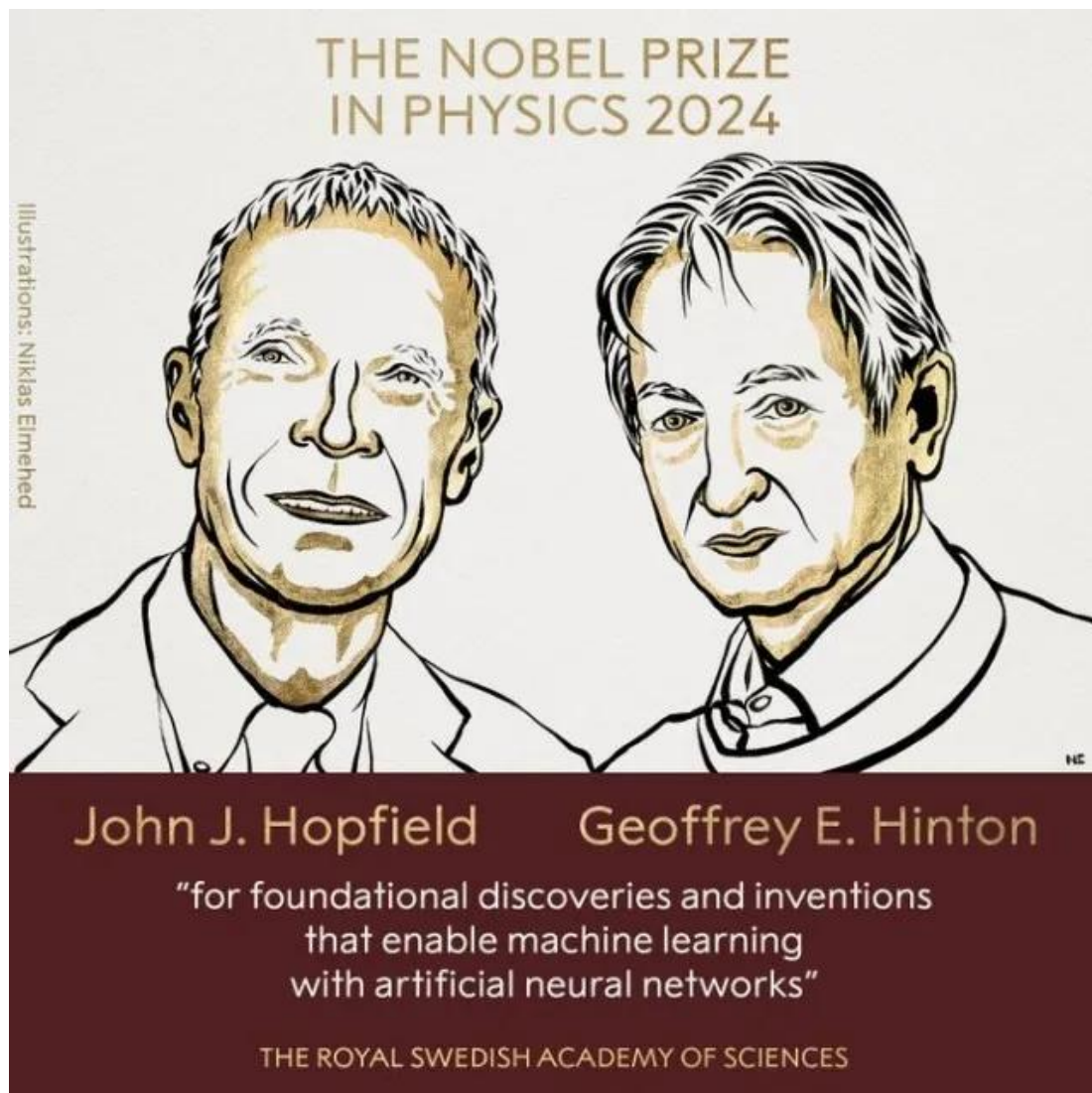
人工智能历史大事件



人工智能历史大事件

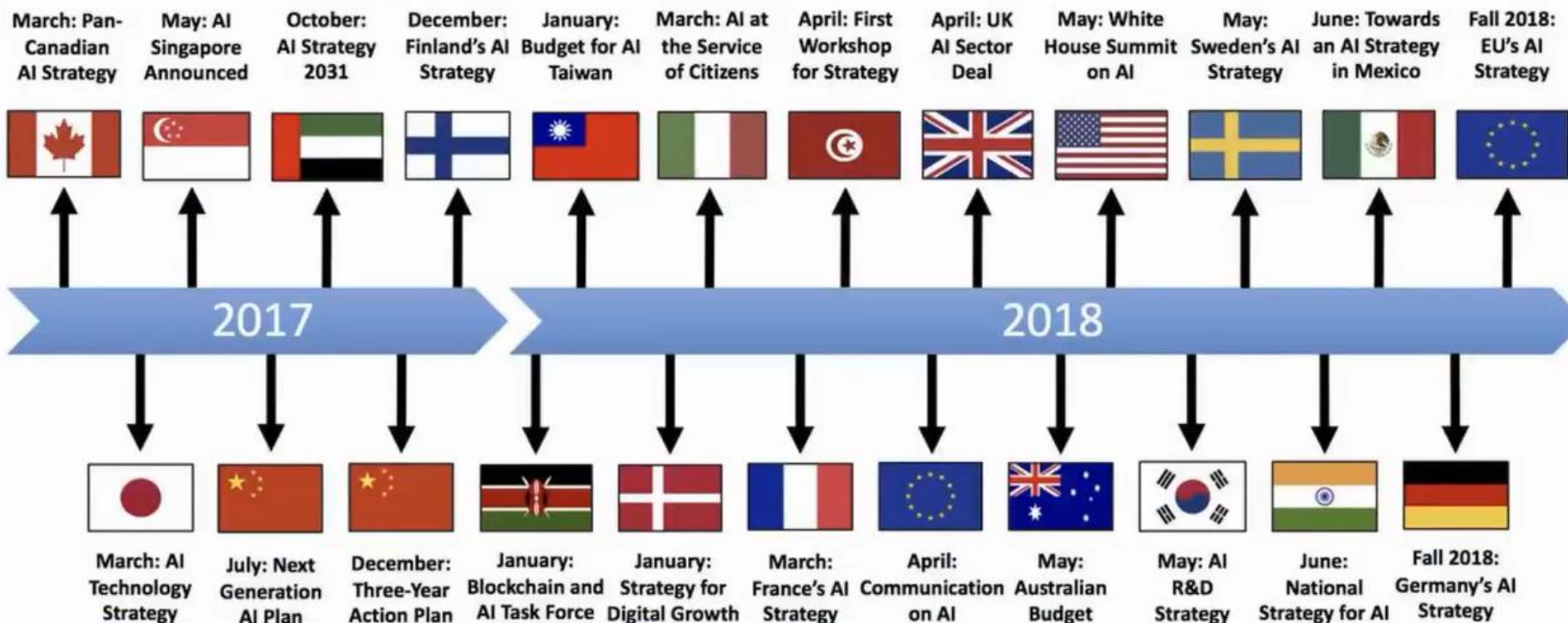
■ 北京时间10月8日下午，2024年诺贝尔物理学奖揭晓。

科学家约翰·霍普菲尔德 John J. Hopfield 和杰弗里·辛顿 Geoffrey E. Hinton 获奖，表彰他们在使用**人工神经网络**进行机器学习的基础性发现和发明。



人工智能的政策支持

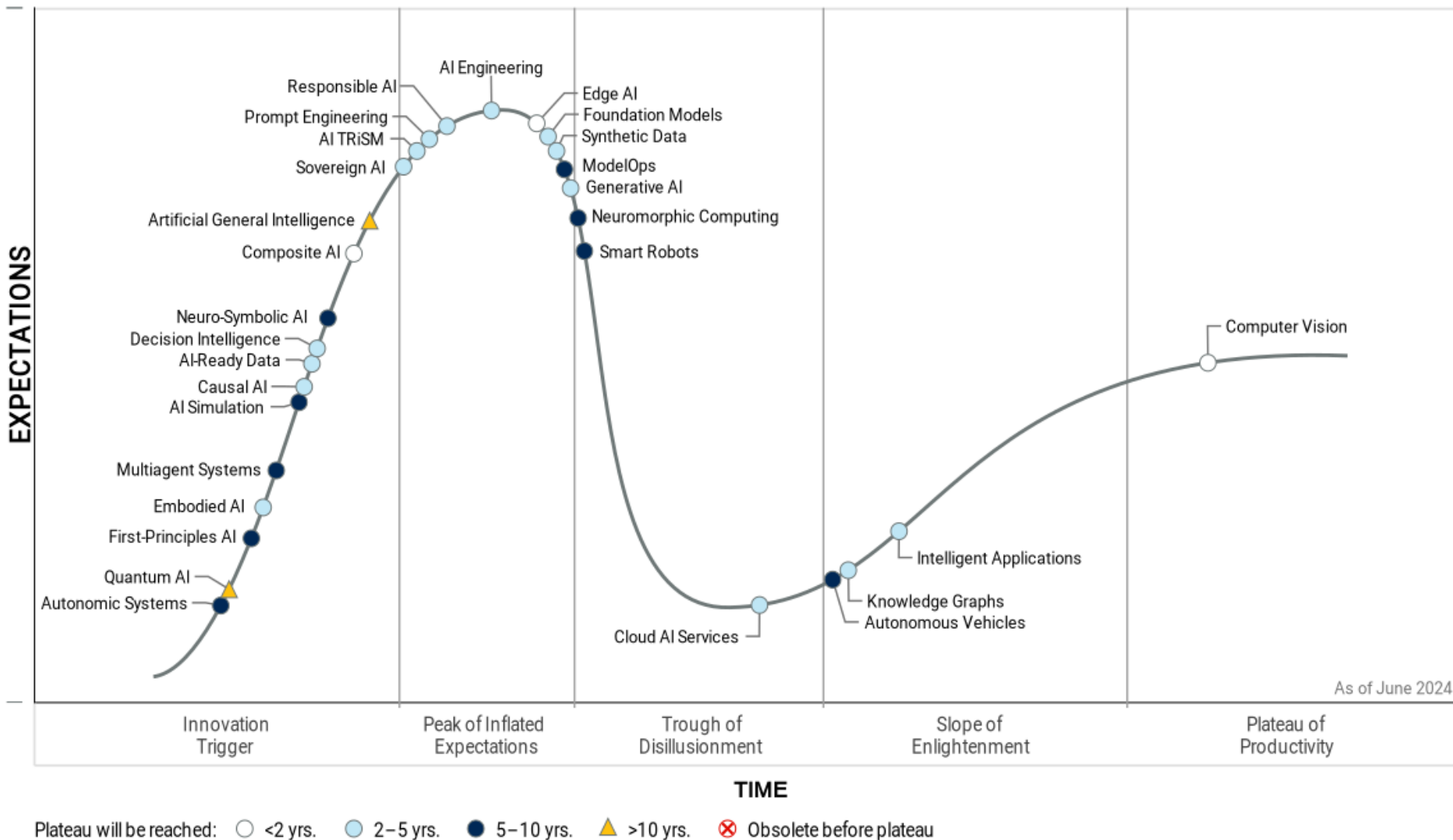
Artificial Intelligence Strategies



人工智能的技术趋势

■ Gartner发布2024年人工智能技术成熟度曲线

Hype Cycle for Artificial Intelligence, 2024



人工智能安全的研究背景

问题挑战

可用性

- 异常输入
- 物理扰动
- 恶意破坏

完整性

- 恶意植入
- 模型篡改
- 数据伪造

鲁棒性

- 对抗样本
- 投毒攻击
- 后门攻击

隐私性

- 模型窃取
- 模型反演
- 成员隐私

深度学习隐私性:是指在数据原始信息没有发生直接泄露的情况下，深度学习模型计算过程产生的信息不会间接暴露训练集数据、模型参数和预测样例。机器学习算法中的隐私问题严重阻碍了机器学习技术的普适推广。

深度学习鲁棒性:是指现有深度学习方法的有效性不依赖于对数据的高质量要求，当训练集呈现蕴含复杂噪声、异常点入侵、类别不均衡等或当测试集存在人为扰动、与异常输入等问题时，其有效性仍能保证。

研究人工智能安全具有重要意义!

人工智能安全的技术内涵

不可解释性

黑箱模型

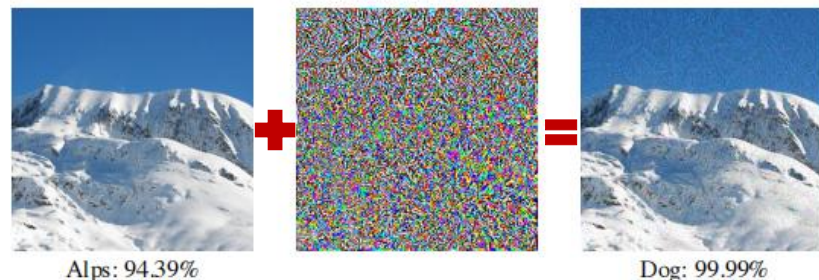


数据强依赖性

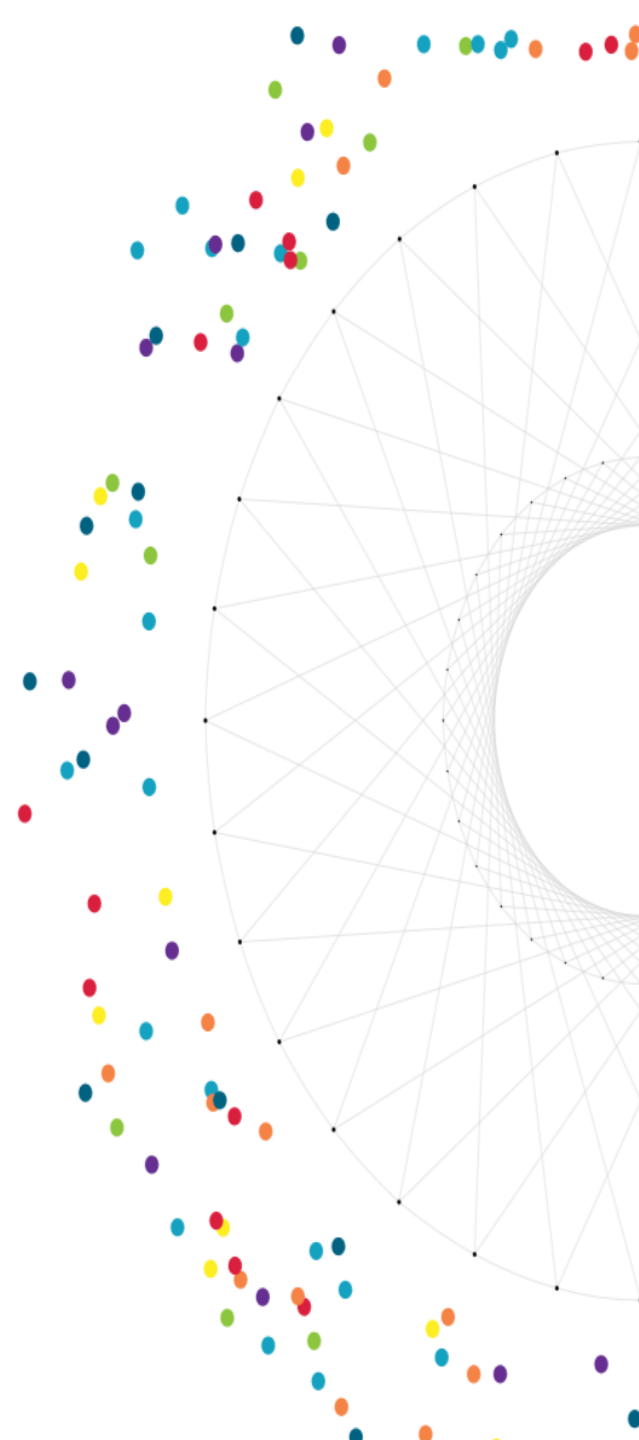
数据偏差和噪声



对抗样本攻击



90年代的媒體：
人工智能會在十年內毀掉社會
現在的人工智能：



Machine Learning in Physical World



Autonomous Driving



Healthcare



Smart City



Malware Classification



Fraud Detection



Biometrics Recognition

Deep fake 动态换脸



Physical Adversarial Attacks Against Sensor Fusion

Goal: we aim to generate physical **adversarial object** against **real-world LiDAR system**.



LiDAR-based perception

Physical Experiments

Adversarial object/benign box
in the middle

Adversarial
Object

Benign
Object



清华最新研究成果：一副眼镜破解 19 款安卓手机人脸识别系统

2021-01-28 22:50:35 来源: IT之家

举报

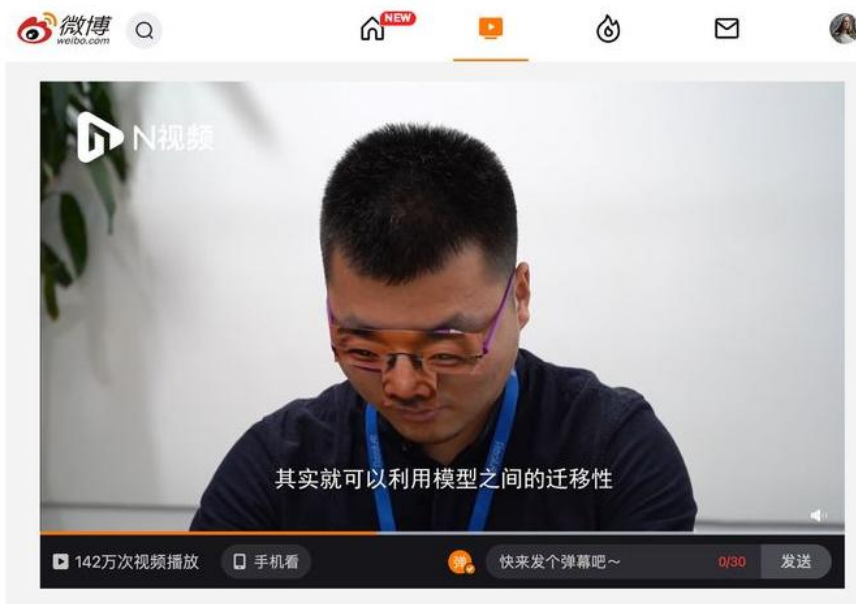


3

分享至



IT之家 1 月 28 日消息 根据清华大学人工智能研究院设立的人工智能企业瑞莱智慧 RealAI 的消息, 研究人员通过对抗样本攻击, 破解了 19 款安卓手机的人脸识别解锁系统。



研究人员表示, 第一步需要拿到攻击目标的照片, 然后用优化算法生成对抗样本, 然后把贴纸裁下来, 贴在眼镜框上就可以了。

官方表示, 他们选取了从低端到高端的 19 款安卓手机和 1 款 iPhone 11 进行了测试, 除了 iPhone 11 之外, 其他的手机都能在 15 分钟之内被破解。另外, 研究人员还用抗样本攻击了几款政务和金融 App 的人脸识别系统, 甚至假冒机主完成了银行开户。



对抗样本是指对原始的输入样本添加精心设计的细微扰动致使机器学习模型输出错误的结果。神经网络因为极易受到对抗样本的威胁而导致其安全问题备受关注。对抗攻击生成的扰动极其微小，人类用肉眼几乎无法察觉，但是对当前的人工智能系统却会造成破坏性的影响，

目前对抗样本已经被证明可以适用到各种视觉分类的任务中，造成重大的安全隐患。例如图1展示了利用对抗样本原理设计的3D眼镜可以破解手机的人脸识别解锁系统，导致财产损失和隐私泄露。由此可见对抗攻击对视觉分类器带来的巨大安全威胁使得研发鲁棒性视觉分类器具有重大现实意义，对人工智能安全的发展具有重要科研价值。

清华大学RealAI

国际背景



2018 年，根据美国《国防授权法案 2019》美国国家人工智能安全委员会（NSCAI）在美国国防部的支持下正式成立。

该委员会的职能在于代表国会研究人工智能与国家安全的相关问题并建言献策，以达到确保美国技术优势的最终目的。

2021年，欧盟提出《人工智能法规草案》



“要加强人工智能发展的潜在风险研判和防范，维护人民利益和国家安全，确保人工智能安全、可靠、可控”



2017年7月国务院发布《**新一代人工智能发展规划**》

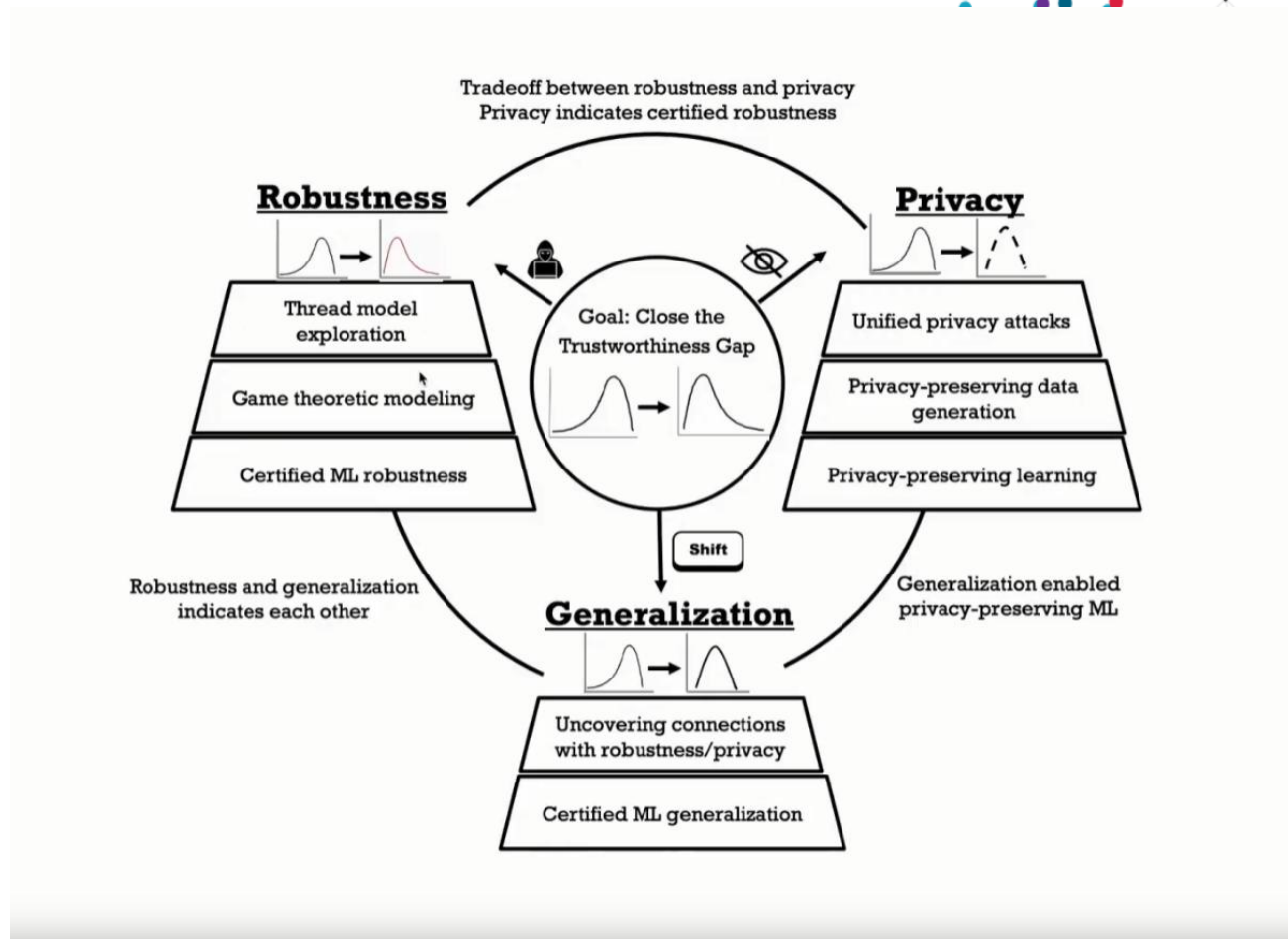
战略目标：围绕核心技术、顶尖人才、标准规范等强化部署，力图在新一轮国际科技竞争中掌握主导权...加强前瞻预防与约束引导，最大限度降低风险，确保人工智能安全、可靠、可控发展。

重点任务：...形成人工智能算法与平台安全性**测试评估的方法、技术、规范和工具集。**

2024年全国网络安全标准化技术委员会发布《**人工智能安全治理框架**》1.0版
目标：鼓励人工智能创新发展为第一要务，以有效防范化解人工智能安全风险为出发点和落脚点，提出了包容审慎、确保安全，风险导向、敏捷治理，技管结合、协同应对，开放合作、共治共享等人工智能安全治理的原则。



AI安全方向分类



02 | Literatures (相关实例)

AI对抗攻防的相关实例

对抗攻防相关实例

■ AI算法：易受“微小”扰动影响



x

“panda”

57.7% confidence

$+ .007 \times$



$\text{sign}(\nabla_x J(\theta, x, y))$

“nematode”

8.2% confidence

$=$



$x +$

$\epsilon \text{sign}(\nabla_x J(\theta, x, y))$

“gibbon”

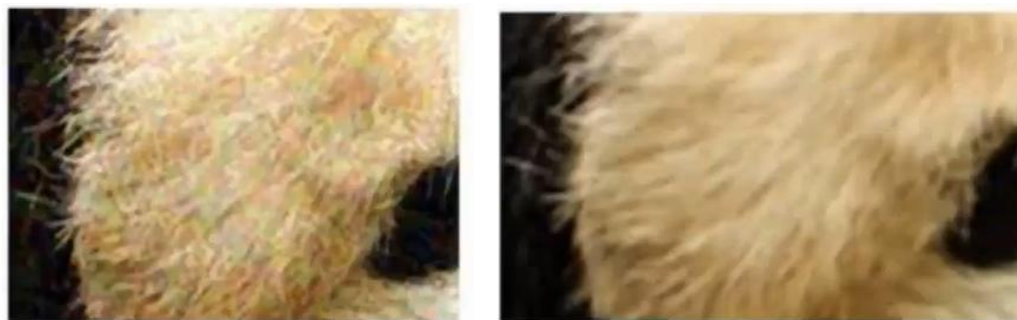
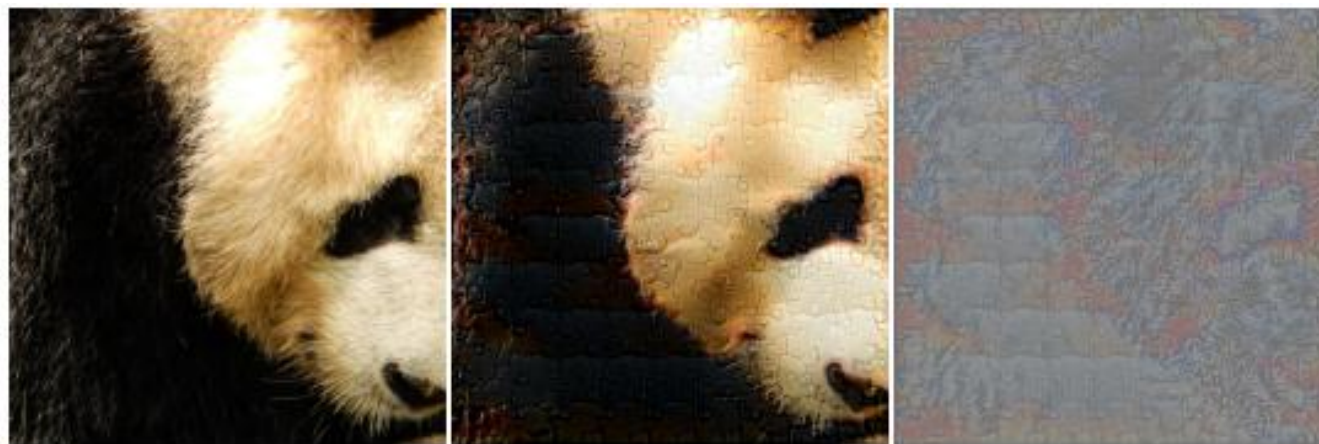
99.3 % confidence

Adversarial perturbation: The noise that is added to the clean image to make it an adversarial example.

[1] Goodfellow, I. J., Shlens, J., and Szegedy, C. Explaining and harnessing adversarial examples. 2015 ICLR.

对抗攻防相关实例

■ “微小” 扰动：人类难以识别



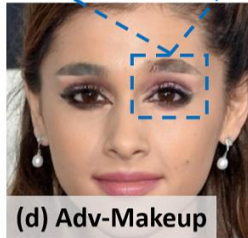
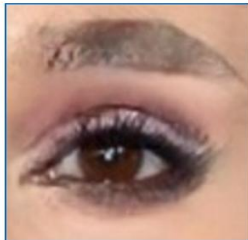
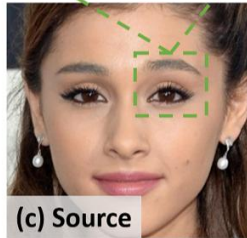
Undistinguishable?

[2] Alexey Kurakin et al. Adversarial Attacks and Defenses Competition, 2017
NeurIPS

How can they be used to compromise machine learning?



Fool the self-driven system



Deceive face recognition

对抗攻防相关实例

■ 单像素攻击：仅改变图片一个像素

AllConv



SHIP
CAR(99.7%)

NiN



HORSE
FROG(99.9%)

VGG



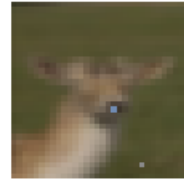
DEER
AIRPLANE(85.3%)

AllConv



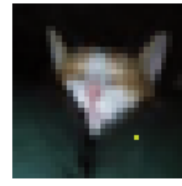
CAR
AIRPLANE(82.4%)

NiN



DEER
DOG(86.4%)

VGG



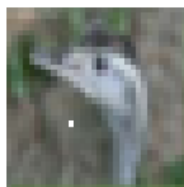
CAT
BIRD(66.2%)



HORSE
DOG(70.7%)



DOG
CAT(75.5%)



BIRD
FROG(86.5%)



DEER
AIRPLANE(49.8%)



BIRD
FROG(88.8%)

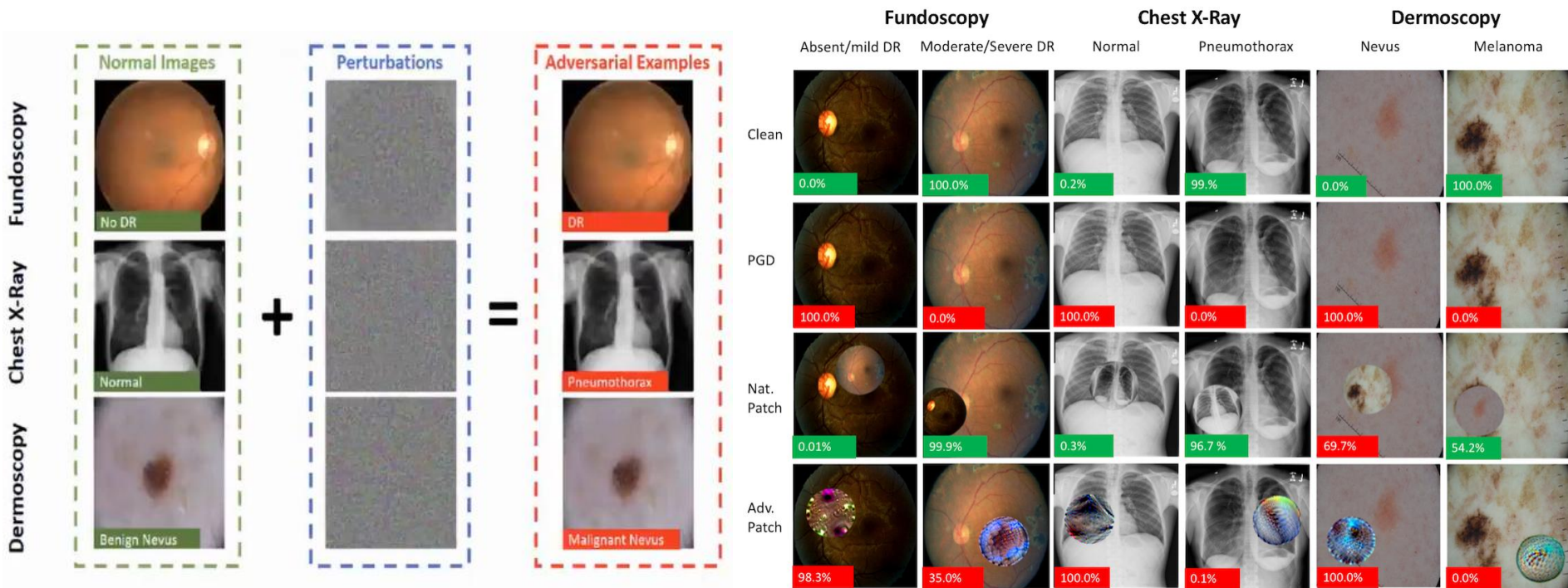


SHIP
AIRPLANE(88.2%)

[3] Jiawei Su et al. One Pixel Attack for Fooling Deep Neural Networks. 2019 TEC.

对抗攻防相关实例

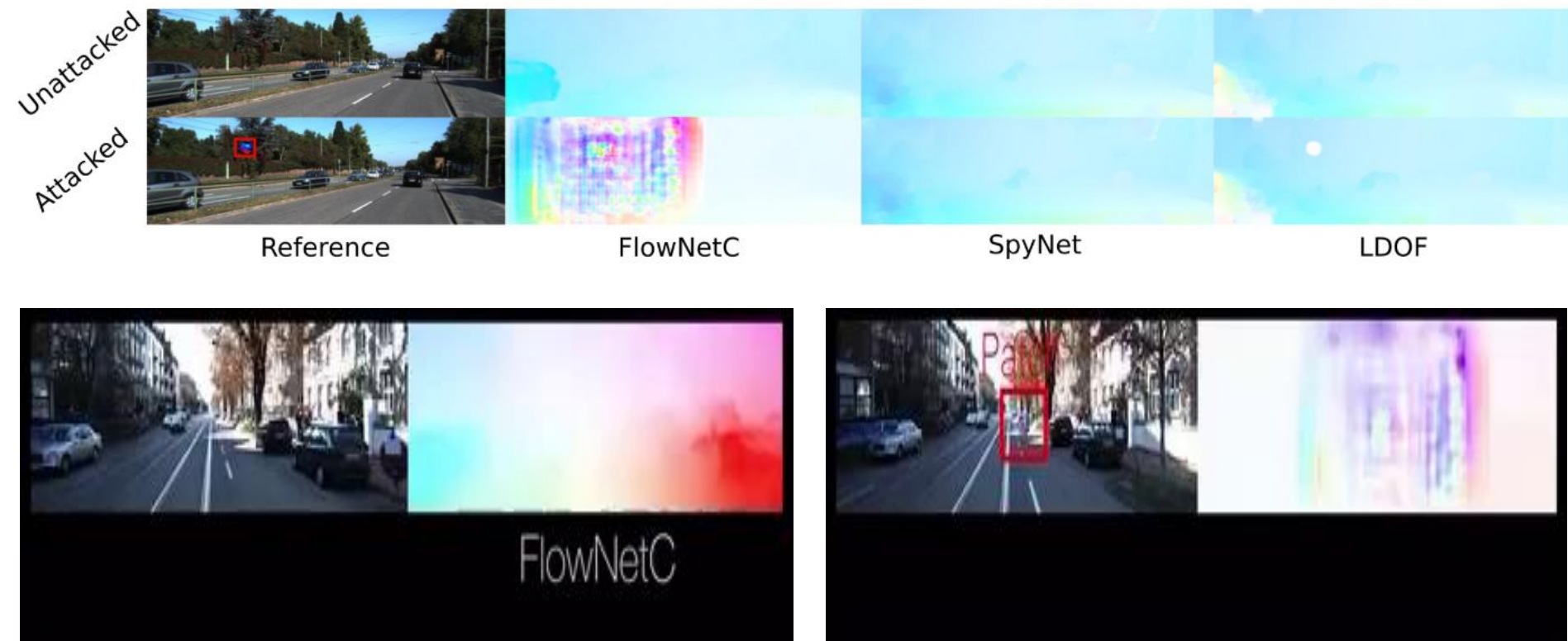
■ 医学领域：医学影像图片识别干扰



[4] Samuel G, Finlayson et al. Adversarial Attacks Against Medical Deep Learning Systems. 2019

对抗攻防相关实例

■ 目标检测：视频中添加扰动——攻击前与攻击后



[5]Anurag Ranjan et al. Attacking Optical Flow.2019

[6]<https://www.youtube.com/watch?v=5n07loiPmdA>

对抗攻防相关实例

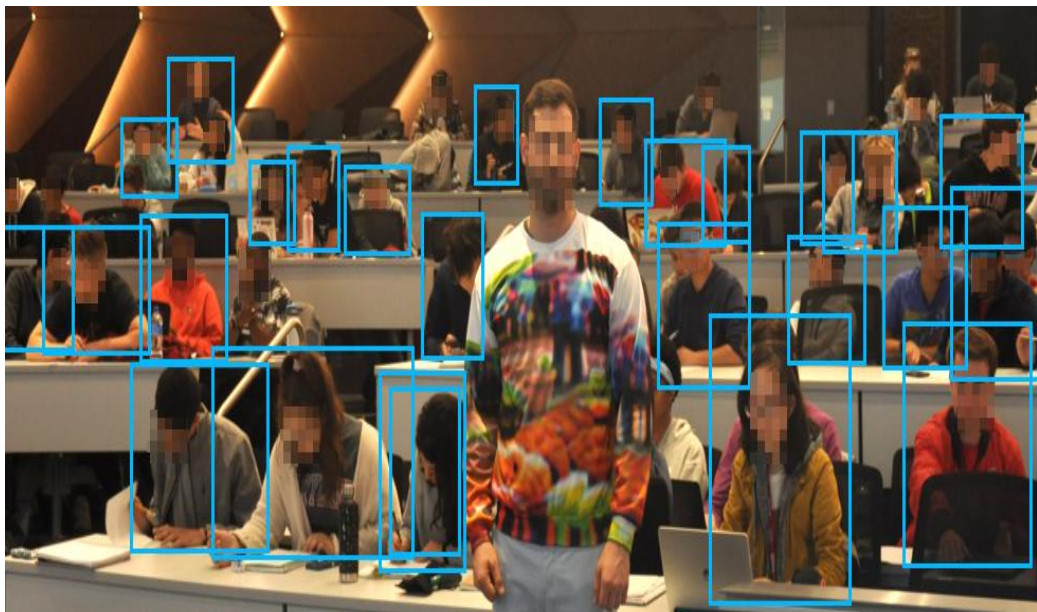
- 自动驾驶的隐患：车载识别系统易受干扰



[7] Yue Zhao et al. Seeing isn't Believing: Towards More Robust Adversarial Attack Against Real World Object Detectors 2019

对抗攻防相关实例

■ 愚弄AI的隐形衣



Original Image



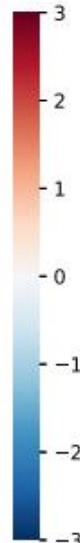
Feature Maps



Patched Image



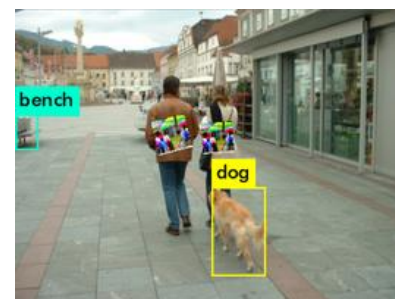
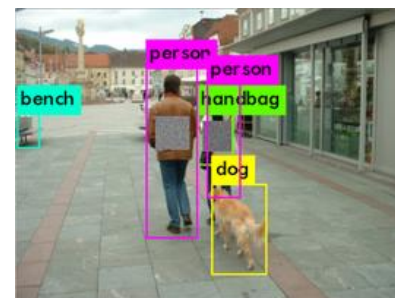
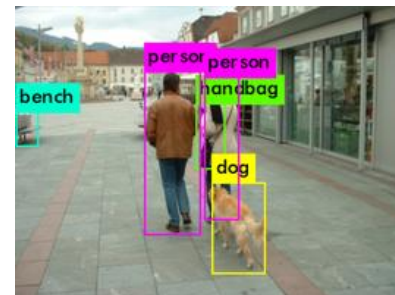
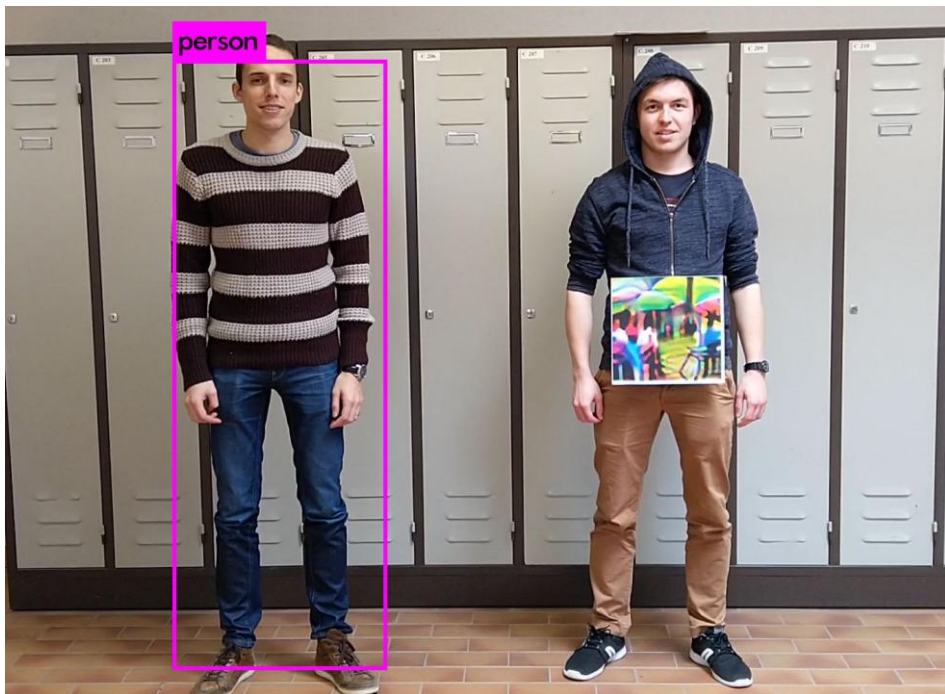
Feature Maps



[8] Zuxuan Wu et al. Making an Invisibility Cloak: Real World Adversarial Attacks on Object Detectors. 2019

对抗攻防相关实例

■ 监控摄像头：“瞒天过海”



[9] Simen Thys and Wiebe Van Ranst, Fooling automated surveillance cameras: adversarial patches to attack person detection.2019

对抗攻防相关实例

- 仿真3D面具：破解人脸识别系统，微信、支付宝、火车站无一幸免



[10]<https://fortune.com/2019/12/12/airport-bank-facial-recognition-systems-fooled/>

对抗攻防相关实例

■ 机器阅读理解：问题预测与情感分类

Article: Super Bowl 50

Paragraph: “Peyton Manning became the first quarterback ever to lead two different teams to multiple Super Bowls. He is also the oldest quarterback ever to play in a Super Bowl at age 39. The past record was held by John Elway, who led the Broncos to victory in Super Bowl XXXIII at age 38 and is currently Denver’s Executive Vice President of Football Operations and General Manager. Quarterback Jeff Dean had jersey number 37 in Champ Bowl XXXIV.”

Question: “What is the name of the quarterback who was 38 in Super Bowl XXXIII?”

Original Prediction: John Elway

Prediction under adversary: Jeff Dean

- 插入干扰性文本（对抗样本）
- 改变机器预测结果：
 - 问题答案预测错误（图1）
 - 情感分类预测相反（图2）

Task: Sentiment Analysis. **Classifier:** Amazon AWS. **Original label:** 100% Negative. **Adversarial label:** 89% Positive.

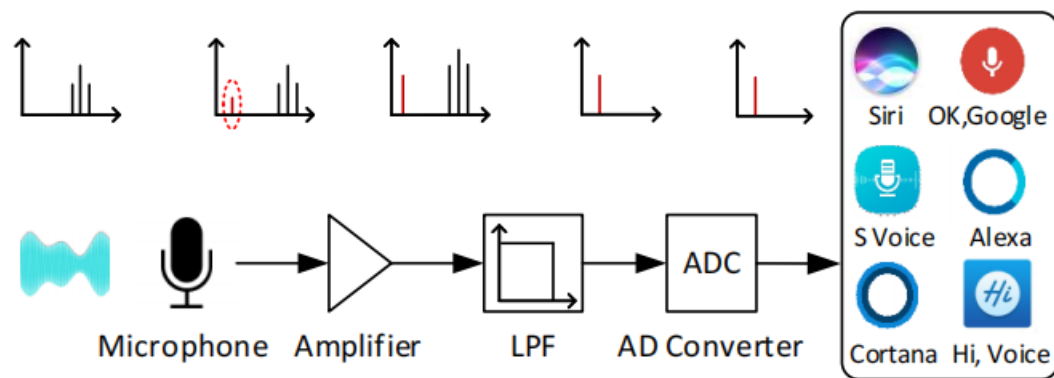
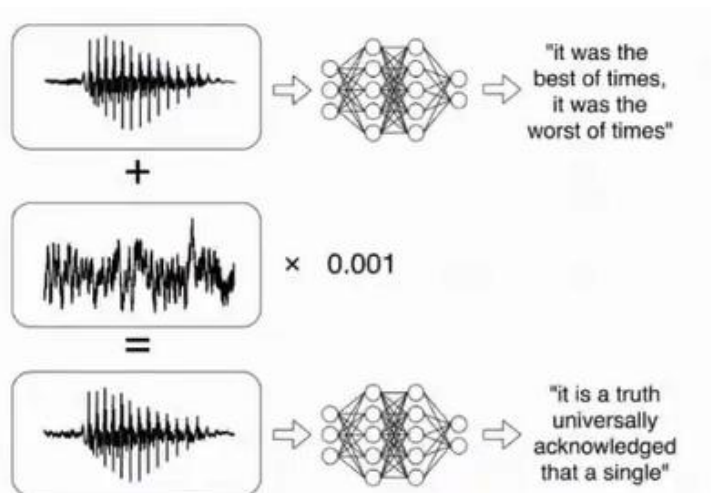
Text: I watched this movie recently mainly because I am a Huge fan of Jodie Foster's. I saw this movie was made right between her 2 Oscar award winning performances, so my expectations were fairly high. Unfortunately **Unf0rtunately**, I thought the movie was terrible **terrib1e** and I'm still left wondering how she was ever persuaded to make this movie. The script is really weak **wea k**.

[11] Jia, R. and Liang, P. Adversarial examples for evaluating reading comprehension systems. 2017

对抗攻防相关实例

■ 语音识别

- 唤醒Siri, 谷歌助手等
- 添加“噪声”
- 改变翻译结果



[12]Zhang G, Yan C, Ji X, et al. Dolphinattack: Inaudible voice commands. 2017

[13]<https://www.youtube.com/watch?reload=9&v=21HF4A3WE4>

03 | Methods (攻防原理)

AI对抗攻防部分算法原理

对抗攻击基本概念

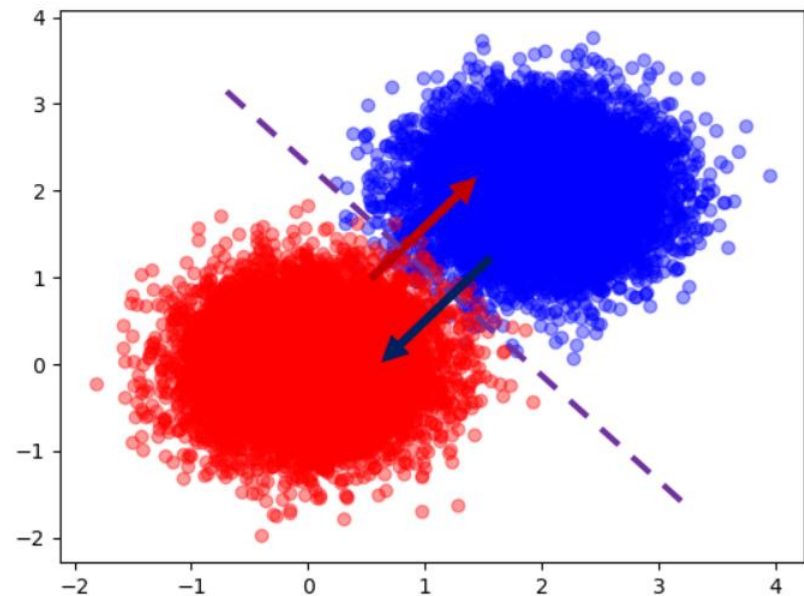
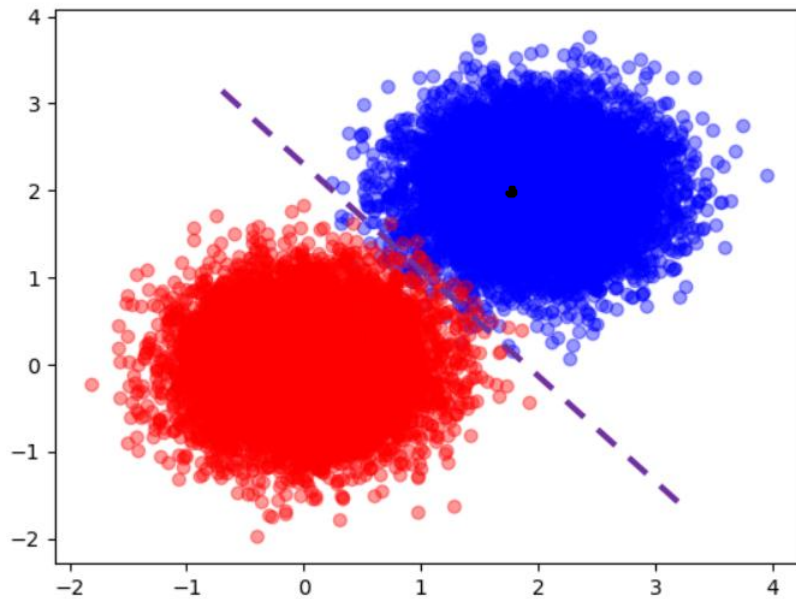
- 现有的对抗攻击方法主要面向对抗样本的生成方法以及最终达到对模型攻击的成功率。而对抗防御主要分为基于对抗样本的检测与提高模型鲁棒性两个方面。

- **深度学习**：深度学习是一种深层模型，利用多层非线性变换进行特征提取；由低层特性抽取出高层更抽象的表示。主要类型：循环神经网络、深度置信网络、卷积神经网络等。
- **对抗攻击**：通过设计一种有针对性的数值型向量从而让机器学习模型做出**误判**，这便被称为对抗性攻击。
- **对抗样本**：人为构造的样本。通过对正常样本 x 添加难以察觉的**扰动** η ，使得分类模型 f 对新生成的样本 x' 产生错误的分类判断。新生成的对抗样本为： **$x' = x + \eta$** 。
同时：

$$f(x') \neq f(x), ||x' - x|| \leq \epsilon$$

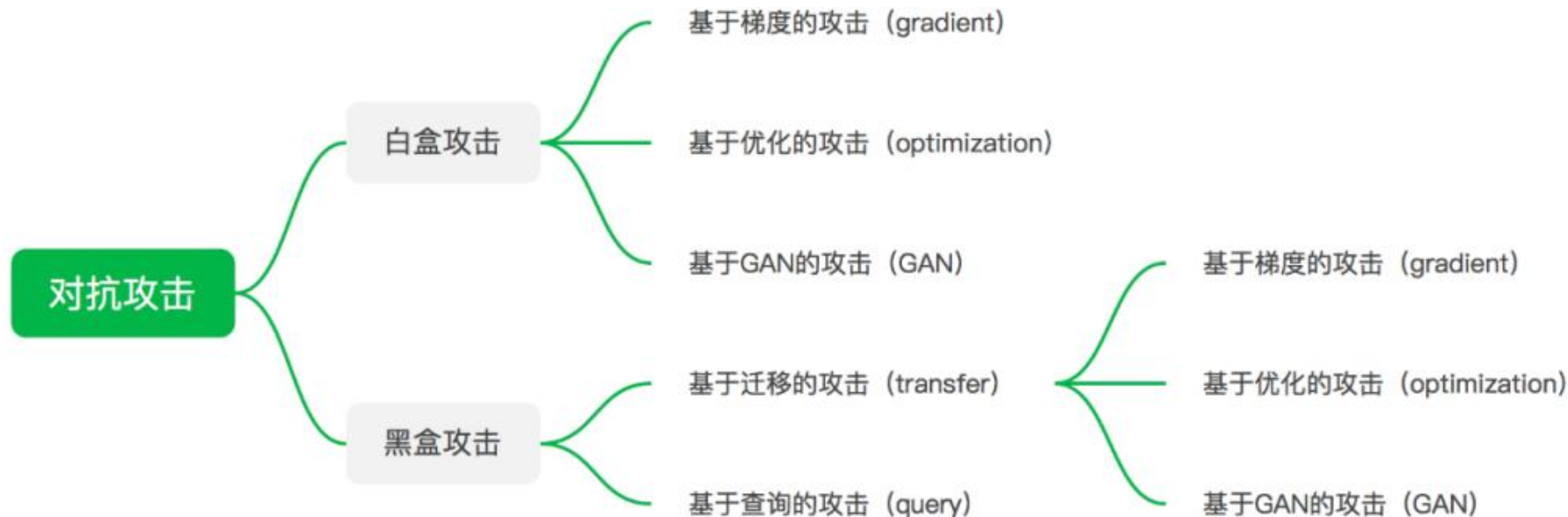
- **可迁移性**：类型有：在同一数据集训练的不同模型之间的可迁移性、在不同机器学习技术之间的可迁移性、执行不同任务的模型之间的可迁移性。影响因素：模型类型、对抗样本的攻击力、非目标攻击比目标攻击更容易迁移、数据统计规律。
- **鲁棒性（健壮性）**：控制系统在一定(结构，大小)的参数摄动下，维持其它某些性能的特性。

4.3 Why do they happen(intuitive)



对抗攻击方法分类

- 根据攻击者对目标模型（被攻击的模型）的了解程度不同，已有的对抗攻击大致可以分为两类:白箱攻击和黑箱攻击。



- **白盒攻击**：当算法模型与训练数据都被攻击者所掌握，并且攻击者在此基础上进行的针对性攻击称为白盒攻击。但是很显然，算法模型与训练数据一般都是商业机密，想要获得这些信息是**高成本**的，也是不太现实的，所以白盒攻击的实用价值受到很大限制。
- **黑盒攻击**：在对算法模型的结构和参数**一无所知**，甚至相关训练数据也一无所知的情况下，进行攻击的过程称为黑盒攻击。这种情况跟工业界遇到的实际场景较为符合，是工业界关注的重点，也是学术界研究的重点方向。

经典攻击方法

■ FGSM: 快速梯度攻击

原理：计算关于输入的损失函数的梯度，将一个选定的小的常数与梯度的符号向量相乘来产生一个小的扰动。

- ①在白盒环境下，通过求模型对输入的导数
- ②用符号函数得其梯度方向
- ③乘以一个步长，得到“扰动”
- ④攻击样本为原输入加“扰动”

$$\eta = \epsilon \text{sign}(\nabla_x J(\theta, x, y)) \longrightarrow x' = x + \eta$$

其中 η ：生成的扰动、 ϵ ：控制扰动大小的参数、 x ：原始样本、 x' ：生成的对抗样本、 y ：目标标签、 θ ：网络参数、 J ：损失函数、 $\nabla J(\theta, x, y)$ ：相对于输入 x 损失函数的一阶导数、 sign ：符号函数(大于0时取1，小于0时取-1)



x

“panda”

57.7% confidence

+ .007 ×



$\text{sign}(\nabla_x J(\theta, x, y))$

“nematode”

8.2% confidence

=



$x + \epsilon \text{sign}(\nabla_x J(\theta, x, y))$

“gibbon”

99.3 % confidence

4.3 Why do they happen

FAST GRADIENT SIGN METHOD (FGSM) (Goodfellow, Shlens et al. 2014)

Adversarial Perturbation: $\eta = \epsilon \text{sign}(\nabla_x J(\theta, x, y))$

$J(\theta, x, y)$: the cost function to train the neural network(θ : the parameters of a model; x : the input to the model; y : the targets associated with x)

∇ : the gradient of the cost function (the required gradient can be computed efficiently using backpropagation)

ϵ : small number

$$\text{sgn}(x) := \begin{cases} -1 & \text{if } x < 0, \\ 0 & \text{if } x = 0, \\ 1 & \text{if } x > 0. \end{cases}$$

Adversarial example: $\hat{x} = x + \eta$

$$\eta = \epsilon \text{sign}(\nabla_x J(\theta, x, y))$$

$$\hat{x} = x + \eta$$

Q1: Why is the formula related to gradient of the cost function?

A: To increase model error $\rightarrow$ increasing the cost function.

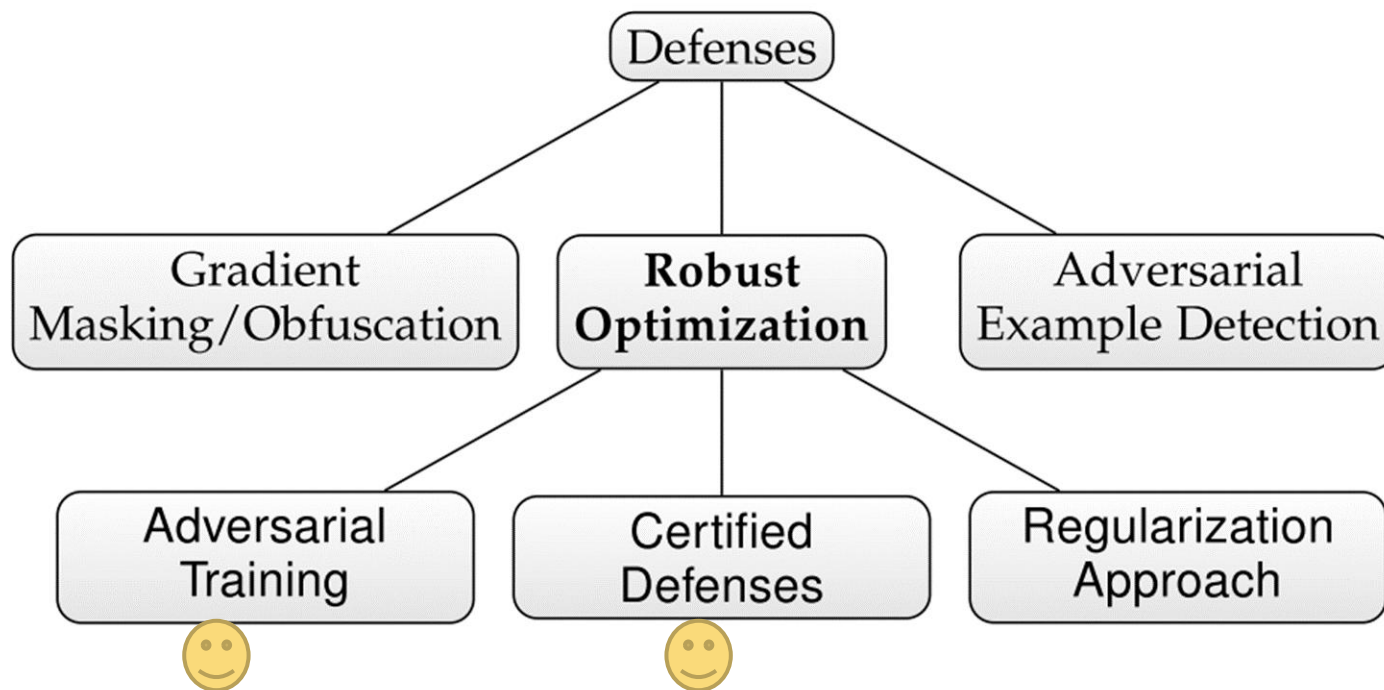
Q2: It looks like Gradient Descent Optimization. Isn't it?

$$W_{ij}^{(l)} = W_{ij}^{(l)} - \alpha \frac{\partial}{\partial W_{ij}^{(l)}} J(W, b)$$

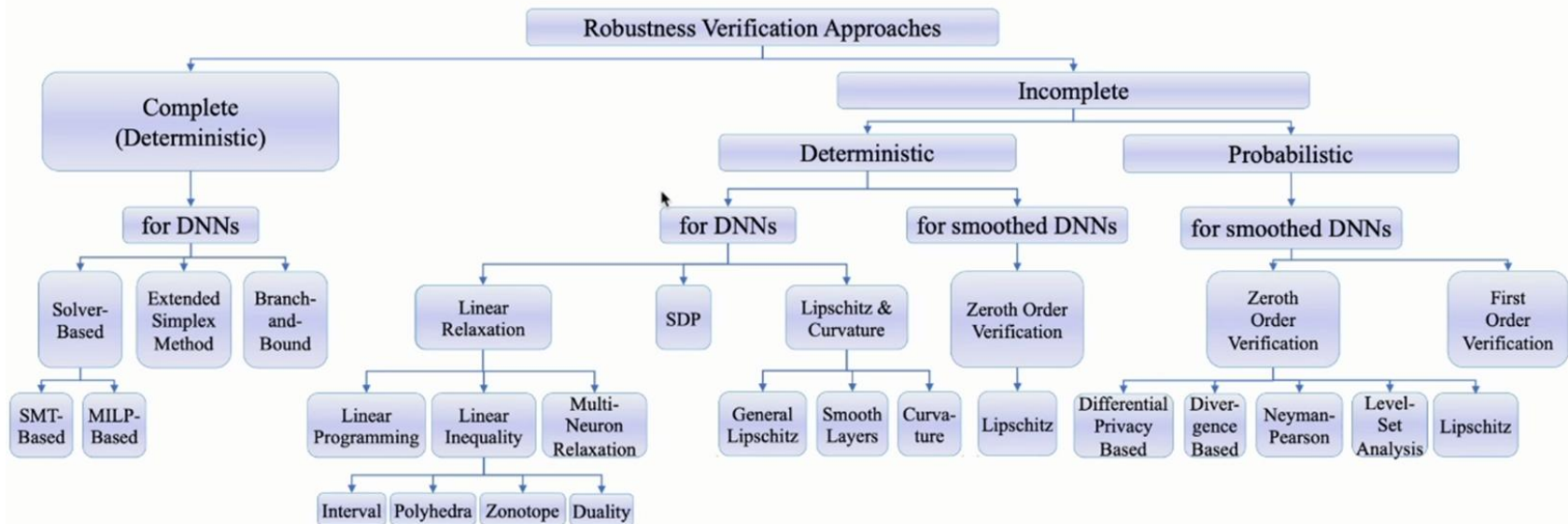
$$b_i^{(l)} = b_i^{(l)} - \alpha \frac{\partial}{\partial b_i^{(l)}} J(W, b)$$

对抗样本如何帮助改善机器学习系统?

● 1) 改善模型的鲁棒性



Certified Robustness For ML



<https://sokcertifiedrobustness.github.io/>

对抗训练**Adversarial Training**

An Optimization View on Adversarial Robustness (Madry, Makelov et al. 2018)

$$\min_{\theta} \rho(\theta), \quad \text{where} \quad \rho(\theta) = \mathbb{E}_{(x,y) \sim \mathcal{D}} \left[\max_{\delta \in \mathcal{S}} L(\theta, x + \delta, y) \right]$$

经典防御方法

■ 提高模型鲁棒性的秘密武器——对抗训练

Szegedy在14年的ICLR中提出了对抗样本的概念。对抗样本可以用来攻击和防御，而对抗训练其实是“对抗”家族中防御的一种方式，其基本原理为：通过添加扰动构建对抗样本，喂入模型一同训练，提高模型遇到对抗样本时的鲁棒性，同时一定程度也能提高模型的表现和泛化能力。

对抗样本一般需要具有两个特点：

①相对于原始输入，所添加的扰动是微小的 ②能使模型犯错

对抗训练的公式如下：
$$\min_{\theta} \rho(\theta), \quad \text{where} \quad \rho(\theta) = \mathbb{E}_{(x,y) \sim \mathcal{D}} \left[\max_{\delta \in \mathcal{S}} L(\theta, x + \delta, y) \right]$$

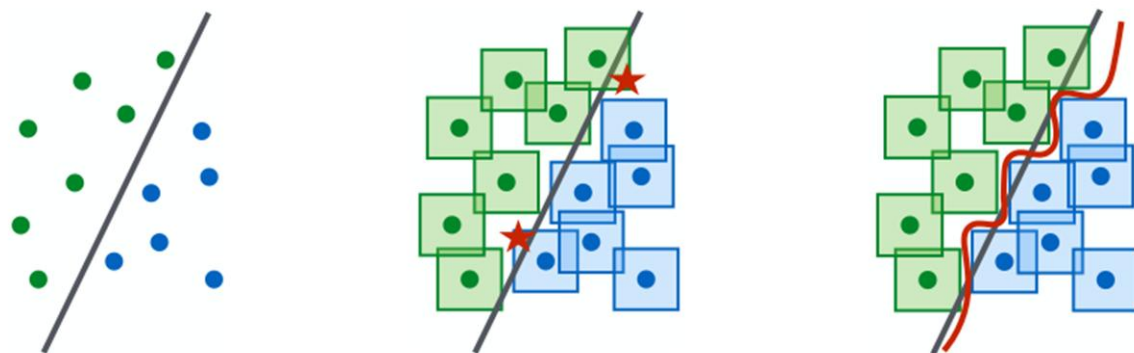
该过程可以分为两步：

内部的max过程：寻找让模型犯错最大的扰动；

外部的min过程：寻找整体损失最小的参数。

对抗训练的作用：

1. 提高模型应对恶意对抗样本时的鲁棒性。
2. 作为一种正则化方式(regularization)，减少过拟合(overfitting)，提高泛化能力。

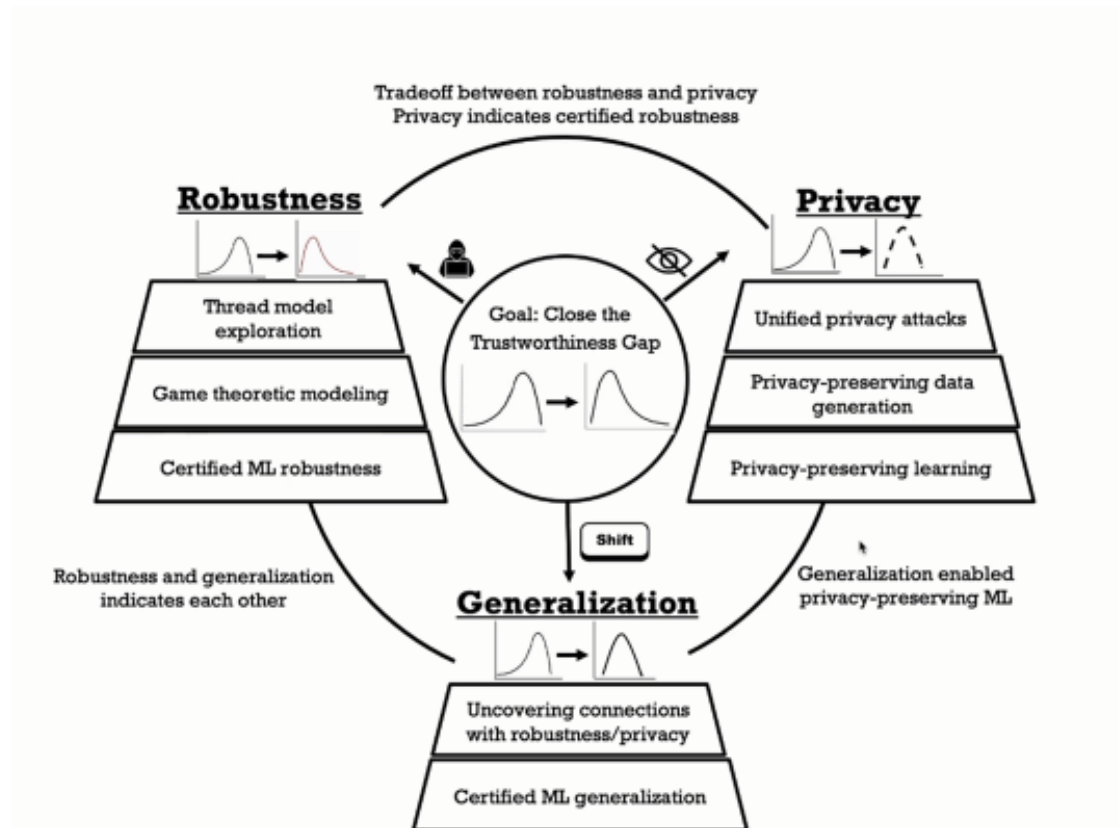


CIFAR10

	Simple	Wide	Simple	Wide	Simple	Wide
Natural	92.7%	95.2%	87.4%	90.3%	79.4%	87.3%
FGSM	27.5%	32.7%	90.9%	95.1%	51.7%	56.1%
PGD	0.8%	3.5%	0.0%	0.0%	43.7%	45.8%
(a) Standard training			(b) FGSM training			(c) PGD training

1.4 How to use adversarial examples to improve machine learning?

●2) Understand what intelligence is and how the network works



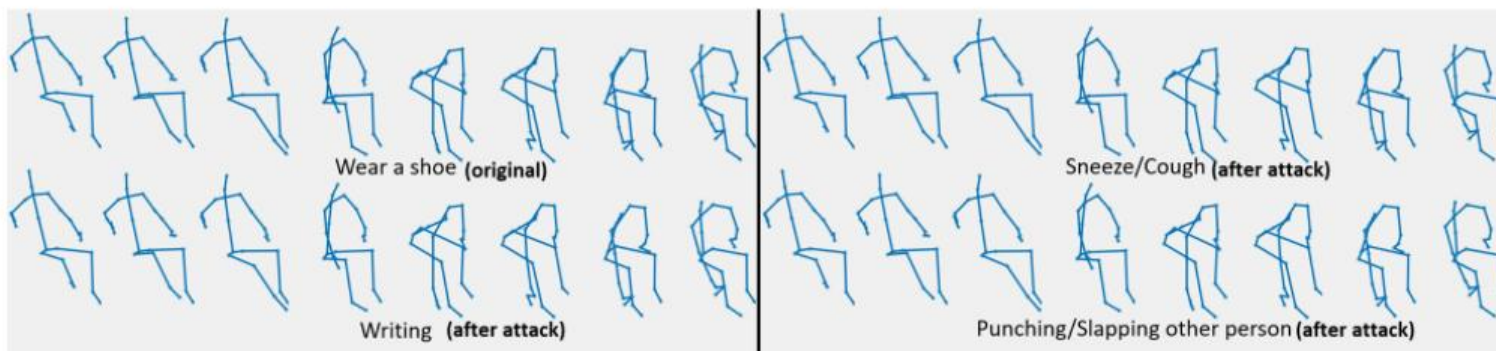
The Adversarial Vulnerability of Visual Classifier and Potential Solutions

The background of the slide features several overlapping geometric shapes in two shades of blue. A light blue shape is positioned behind the title text. To the right, there are darker blue shapes, including a large triangle and a parallelogram, which overlap each other and the light blue shape.

已有工作

Attack Setting	Information Access		Back Propagation	Query
White-box	x Train Dataset	Weights, logit layer, softmax layer, algorithm, hidden layers, gradient, loss functions, activation y Class Scores	Full Access	Fast Unlimited
Black-box	x	Black-Box y	No Access	Restriceted

针对行为识别模型的对抗攻击与防御



■行为识别：一种重要的计算机视觉任务和时间序列数据，应用于自动驾驶、监控安防、人机交互和医疗检测等与人生命安全息息相关的场景。

■当前时序数据，尤其是针对行为识别的对抗鲁棒性研究较少

■我们关注3D（骨架）行为识别模型：相比于视频数据，3D骨架数据对于光照，视角，遮挡等具有更好的鲁棒性

挑战

- 对抗攻防在骨架行为识别领域缺少相关研究
- 扰动空间的自由度低
 - 每帧样本不到100个自由度，限制了攻击空间
- 感知敏感性 (Perceptual Sensitivity)
 - 动力学敏感性
 - 骨架拓扑结构敏感性
- 非流形对抗样本假设 (Off-manifold Adversarial Samples)
 - 之前的对抗样本主要假设对抗样本位于数据流形之外。不同于数字数据（图片、视频），骨骼数据是现实世界中真实存在的数据，其中非流形对抗样本容易被感知。

BASAR (black-box attack on skeletal action recognition)

问题:

- 骨架对抗攻击是否是一种现实中真实的安全威胁?
- 骨架对抗样本是否存在于真实世界中?

贡献:

- 第一个针对骨架行为识别的黑盒对抗攻击方法
- 首次验证了流形对抗样本在时序数据上的存在
- 验证了流形对抗样本人类不易察觉, 具有极大的安全威胁

Diao, Yunfeng, Tianjia Shao, Yong-Liang Yang, Kun Zhou, and He Wang. "BASAR: black-box attack on skeletal action recognition." CVPR 2021(CCF-A).

Method:

Existing method: considering the classification during attack

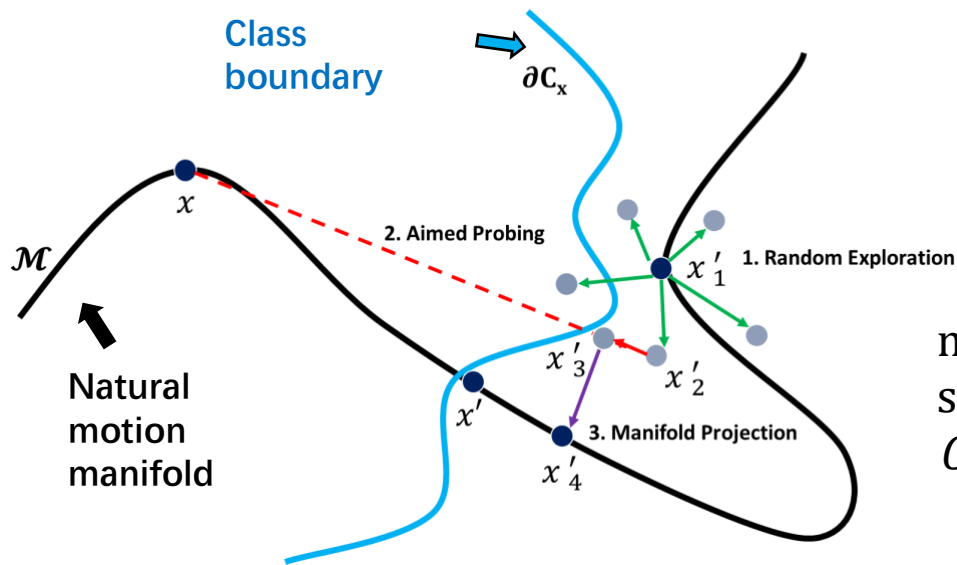
$$\begin{aligned} &\text{minimize } L(\mathbf{x}, \mathbf{x}') \\ &\text{subject to } \mathbf{x}' \in [0, 1]^{m \times n} \\ &C_{\mathbf{x}'} = c \text{ (targeted) or } C_{\mathbf{x}'} \neq C_{\mathbf{x}} \text{ (untargeted)} \quad (1) \end{aligned}$$

Our method: explores the interplay between the classification boundary and the natural motion manifold

$$\begin{aligned} &\text{minimize } L(\mathbf{x}, \mathbf{x}') \\ &\text{subject to } \mathbf{x}' \in [0, 1]^{m \times n}, \mathbf{x}' \in \mathcal{M} \\ &C_{\mathbf{x}'} = c \text{ (targeted) or } C_{\mathbf{x}'} \neq C_{\mathbf{x}} \text{ (untargeted)} \quad (2) \end{aligned}$$

Where $\mathcal{M}$ is a natural pose manifold

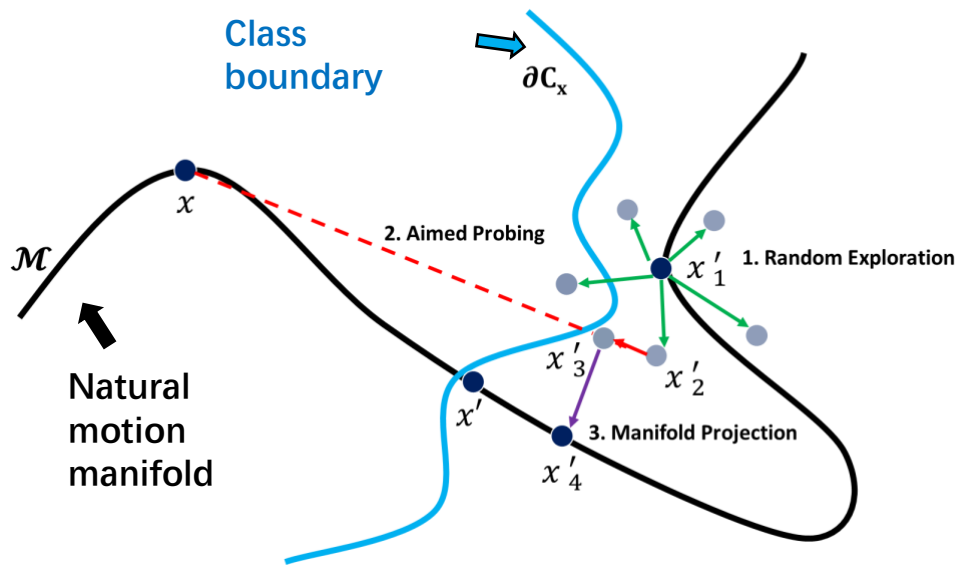
BASAR



minimize $L(\mathbf{x}, \mathbf{x}')$
subject to $\mathbf{x}' \in [0, 1]^{m \times n}$, $\mathbf{x}' \in \mathcal{M}$
 $C_{\mathbf{x}'} = c$ (targeted) or $C_{\mathbf{x}'} \neq C_{\mathbf{x}}$ (untargeted)

Where $\mathcal{M}$ is a natural pose manifold

BASAR



(1) Random Exploration

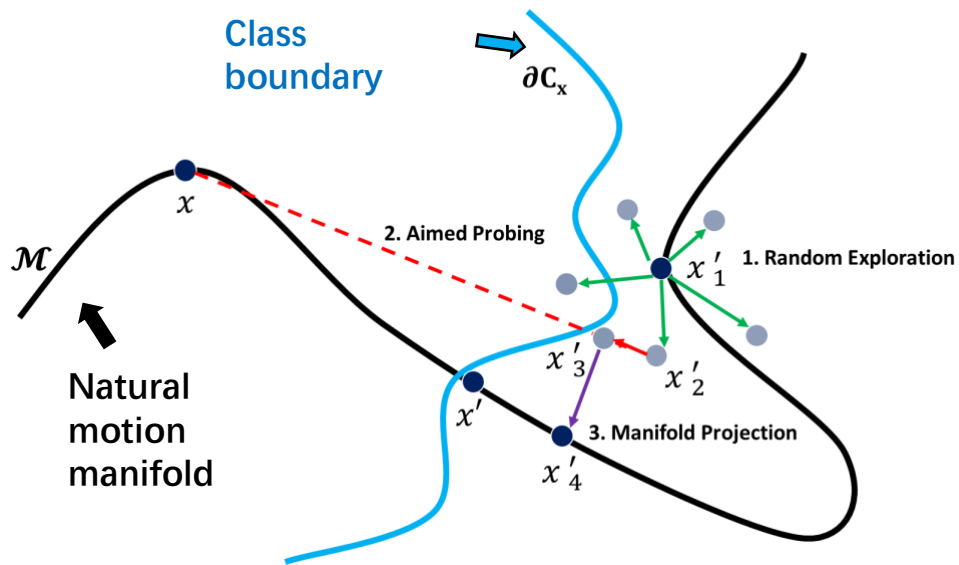
$$\tilde{\mathbf{x}} = \mathbf{x}' + \mathbf{W}\Delta, \text{ where } \Delta = \mathbf{R} - (\mathbf{R}^T \mathbf{d})\mathbf{d}, \mathbf{d} = \frac{\mathbf{x} - \mathbf{x}'}{\|\mathbf{x} - \mathbf{x}'\|},$$

$$\mathbf{R} = \lambda \frac{\mathbf{r}}{\|\mathbf{r}\|} \|\mathbf{x} - \mathbf{x}'\|, r \in N(0, \mathbf{I}), \quad (3)$$

(2) Aimed Probing

$$\tilde{\mathbf{x}} = \mathbf{x}' + \beta(\mathbf{x} - \mathbf{x}'),$$

BASAR



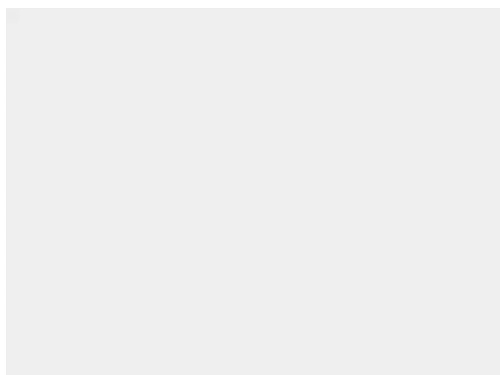
(3) Manifold Projection

minimize $L(\mathbf{x}, \mathbf{x}') + wL(\ddot{\mathbf{x}}, \ddot{\mathbf{x}}')$
 subject to $B'_i = B_i$ and $\theta_i^{\min} \leq \theta'_i \leq \theta_i^{\max}$
 $C_{\mathbf{x}'} = c$ (targeted) or $C_{\mathbf{x}'} \neq C_{\mathbf{x}}$ (untargeted),

Comparison

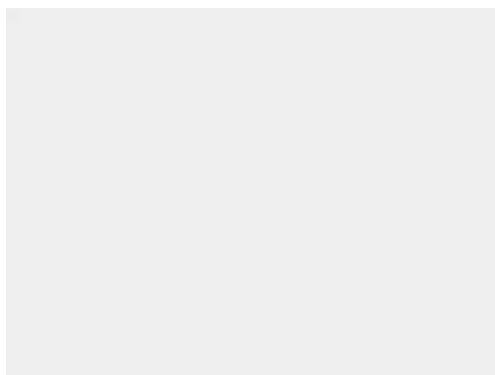
Dataset: NTU

Attacked Sample



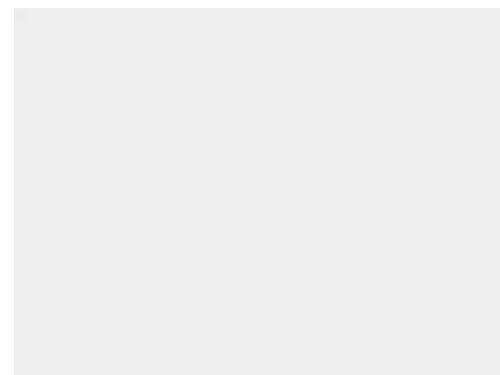
Ground Truth:
Sneeze

BA (Brendel, Rauber et al. 2017)



After Attack:
Vomiting

Ours



After Attack:
Vomiting

Comparison

Dataset: HDM05

Attacked Sample



Ground Truth:
Stand up

(Wang, He et al. 2019)
SMART (White box)



After Attack:
Deposit

Ours

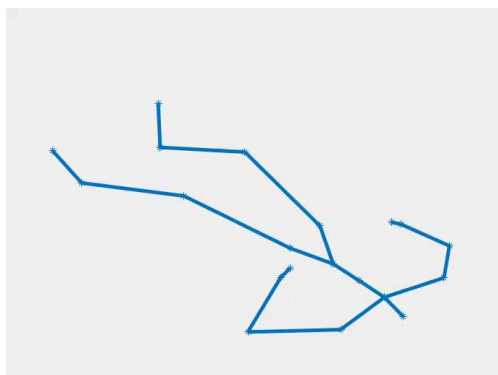


After Attack:
Sneak

Comparison

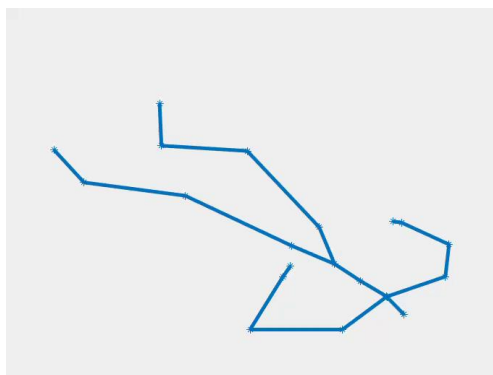
Dataset: HDM05

Attacked Sample



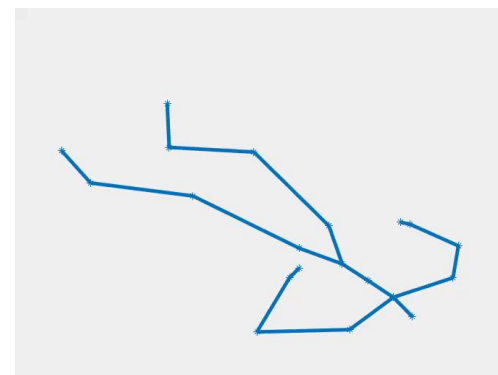
Ground Truth:
Stand up from lying

MTS (Harford, Karim et al. 2020)



After Attack:
Sit and throw top

Ours

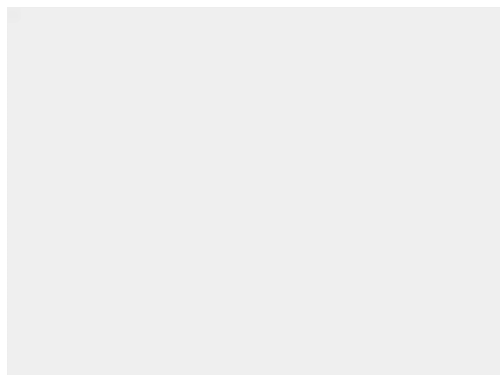


After Attack:
Punch right front

Ablation: On-manifold vs Off-manifold sample

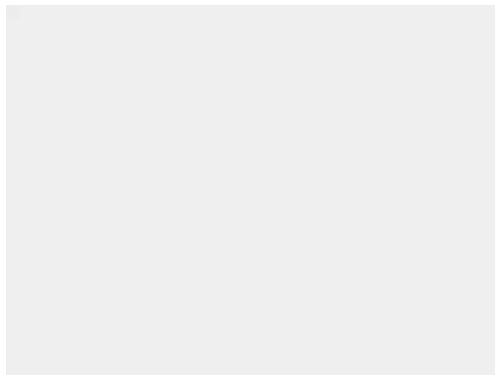
Dataset: NTU

Attacked Sample



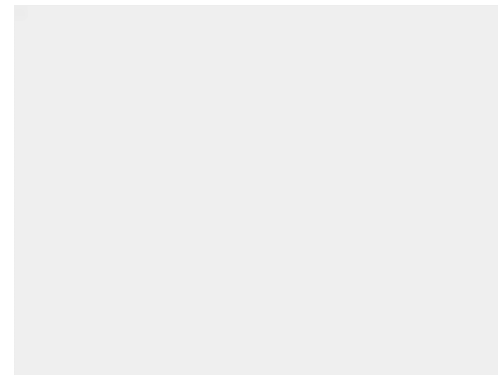
Ground Truth:
Eat meal

Off-manifold sample



After Attack:
Type on a keyboard

On-manifold sample



After Attack:
Play with phone

对抗防御

Mixed On-manifold Adversarial Training

- 针对骨架行为识别的对抗防御方法较少
- 存在的防御方法直接用于骨架行为识别任务效果不佳
 - 对抗训练
 - 随机平滑

Yunfeng Diao, He Wang, Tianjia Shao, Yong-Liang Yang, Kun Zhou, David Hogg, Meng Wang, Understanding the Vulnerability of Skeleton-based Human Activity Recognition via Black-box Attack, Pattern Recognition (CCF-B,中科院一区) 2024.

Existing Work

对抗训练策略:

$$\min_{\theta} E_{\mathbf{x}} [\max_{\sigma \in S} L(\theta, \mathbf{x} + \sigma, y)]$$

L : classification loss function,

σ : perturbation, needs to be computed by a predefined attacker

S : perturbation set

随机平滑策略:

$$\arg \max_{y \in \mathbf{y}} p(g_{\theta}(\mathbf{x} + \sigma) = y) \text{ where } \sigma \sim \mathbf{N}$$

两种策略对于对抗样本分布的假设都过于简化!

where a perturbation σ is drawn from an isotropic Gaussian.

研究动机

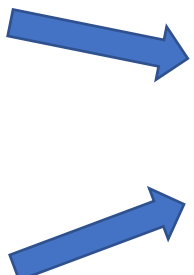
Our assumption: there is a finer structure described by two different distributions in $\mathcal{S}$:
the distributions of **on-manifold** and **off-manifold** adversarial samples

The distributions of **off-manifold** adversarial samples:

more **aggressive**, improving the **robustness**

The distributions of **on-manifold** adversarial samples:

generalization errors, simultaneously improving the
accuracy and robustness



eliminating the **trade-off** between
accuracy and robustness!

Mixed On-manifold Adversarial Training

$$\min_{\theta} \mathbb{E}_{(\mathbf{x}, y) \sim D} [\mu_c \mathcal{L}_c + \mu_{on} \mathcal{L}_{on} + \mu_{off} \mathcal{L}_{off}] \quad (7)$$

where μ_c , μ_{on} and μ_{off} are weights, $\mu_c = 1 - \mu_{on} - \mu_{off}$. The losses are:

$$\mathcal{L}_c = \mathcal{L}(\mathbf{x}, \boldsymbol{\theta}, y) \quad (8)$$

$$\mathcal{L}_{on} = \mathcal{L}(\mathbf{x}'_{on}, \boldsymbol{\theta}, y) \text{ s.t. } \mathbf{x}'_{on} \in [0, 1]^{m \times n}, \mathbf{x}'_{on} \in \mathcal{M} \quad (9)$$

$$\mathcal{L}_{off} = \max \mathcal{L}(\mathbf{x}'_{off}, \boldsymbol{\theta}, y) \text{ s.t. } \mathbf{x}'_{off} \in [0, 1]^{m \times n} \quad (10)$$

实验结果

Methods	BASAR-NoMP			SMART		CIASA		Acc
	$l \uparrow$	$\Delta a \uparrow$	$\Delta B/B \uparrow$	@50	@200	@50	@200	
ST	0.1	0.04	0.66%	3.76%	0.58%	1.69%	0.58%	87.2%
RS	2.80	1.16	19.72%	64.57%	14.33%	62.55%	11.71%	90.4%
MART	6.26	2.14	39.27%	64.84%	52.50%	63.62%	49.33%	70.2%
TRADES	5.25	1.72	30.68%	83.21%	65.83%	81.78%	63.54%	85.8%
MMAT	5.98	2.56	47.98%	85.68%	62.33%	85.31%	60.91%	91.0%
ST	0.70	0.09	1.81%	6.59%	1.47%	6.10%	4.19%	94.4%
RS	6.90	2.59	48.08%	90.35%	39.56%	90.57%	57.44%	93.4%
MART	8.71	3.70	77.26%	88.09%	81.84%	87.70%	80.27%	90.6%
TRADES	10.43	4.19	87.34%	88.83%	84.33%	88.63%	82.90%	90.0%
MMAT	10.30	3.97	79.07%	91.69%	84.79%	91.71%	85.06%	93.8%
ST	7.93	2.00	39.64%	63.29%	22.56%	69.15%	22.56%	94.1%
RS	12.87	3.07	64.18%	79.14%	32.78%	76.71%	24.76%	93.2%
MART	8.46	2.04	40.46%	84.77%	31.25%	83.79%	27.92%	92.1%
TRADES	8.46	2.04	40.44%	84.18%	36.53%	82.62%	25.08%	91.2%
MMAT	13.06	3.46	65.69%	87.36%	48.44%	87.00%	45.15%	93.9%

ST-GCN

MS-G3D

SGN

TASAR: TRANSFER-BASED ATTACK ON SKELETAL ACTION RECOGNITION

Yunfeng Diao¹ **Baiqi Wu**^{1,†} **Ruixuan Zhang**¹ **Ajian Liu**² **Xiaoshuai Hao**³
Xingxing Wei⁴ **Meng Wang**¹ **He Wang**^{5,†}

¹Hefei University of Technology ²Institute of Automation Chinese Academy of Sciences

³Beijing Academy of Artificial Intelligence ⁴Beihang University

⁵UCL Centre for Artificial Intelligence, Department of Computer Science, University College London

`diaoyunfeng@hfut.edu.cn`, `{2021214516, 2020217721}@mail.hfut.edu.cn`, `ajianliu92@gmail.com`

`xshao@baai.ac.cn`, `xxwei@buaa.edu.cn`, `eric.mengwang@gmail.com`, `he_wang@ucl.ac.uk`

迁移对抗攻击

TASAR: Transferable Attack on Skeletal Action Recognition

现存问题:

- 骨骼行为识别在多个领域有重要应用，易受攻击
- 大多针对3D行为识别的对抗攻击聚焦于白盒攻击或查询攻击，缺少针对迁移性攻击的研究
- 已有的针对图片的迁移攻击方法在3D行为识别任务上效果不佳
- 不同替代模型之间的差异巨大

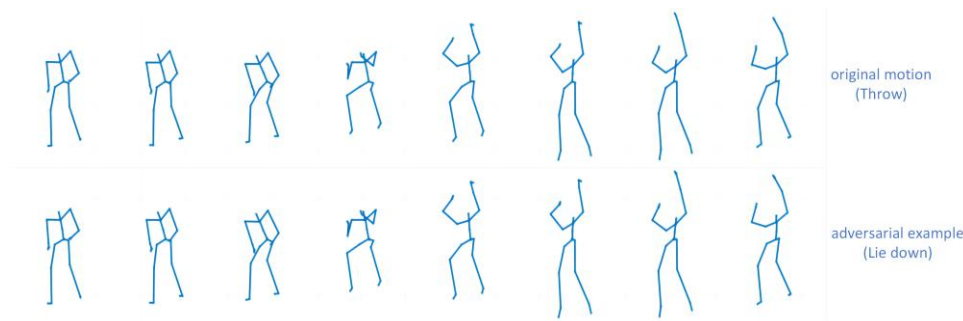


图 对抗样本与原始样本对比

迁移对抗攻击

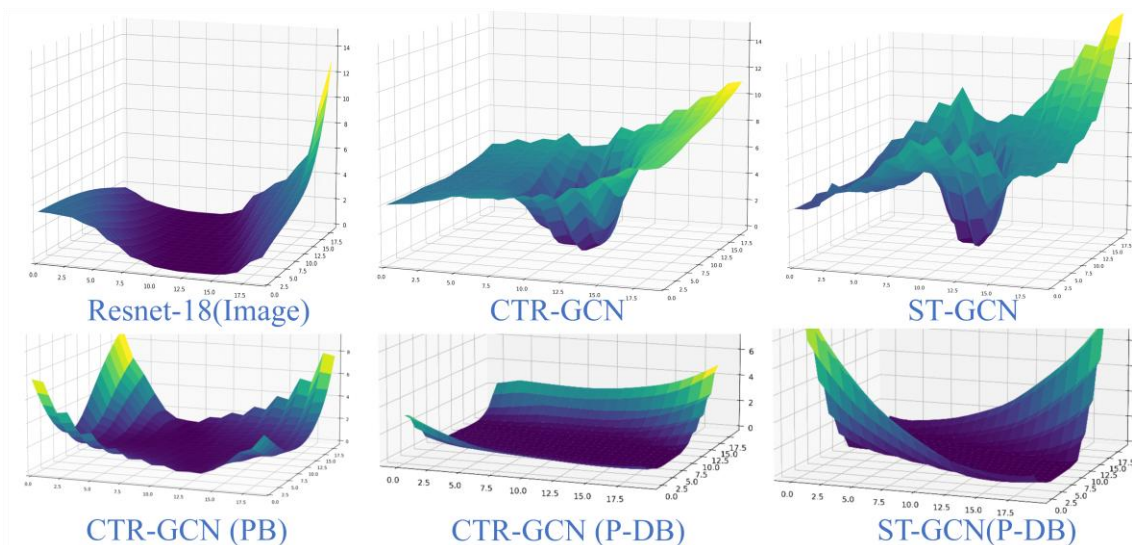
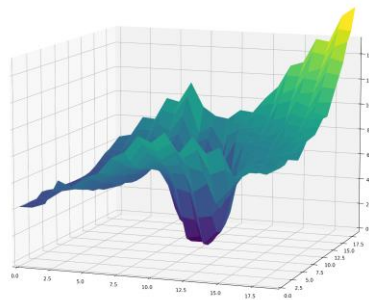
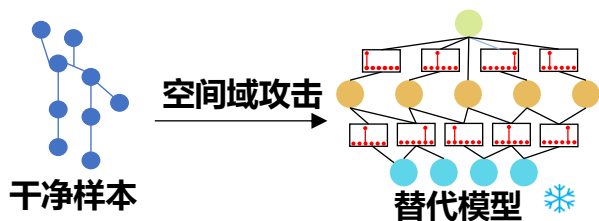


图 模型Loss Landscape

1. 实验结果发现，不同替代模型的**损失景观图平滑性差异巨大**，且图像模型的损失景观图非常平滑。
2. 贝叶斯思想被提出用以平滑模型参数，但**完整采样的时间开销巨大**。
3. 因此，如何**低开销**地构建具备**高迁移性**的替代模型是改善对抗迁移性的重点

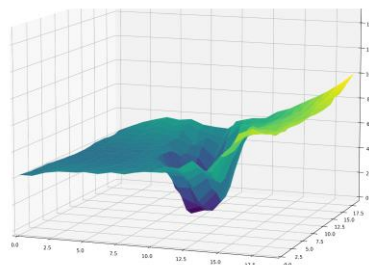
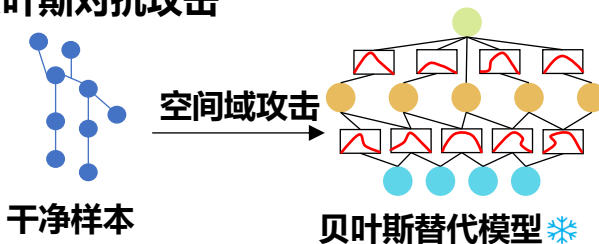
研究现状

确定性对抗攻击



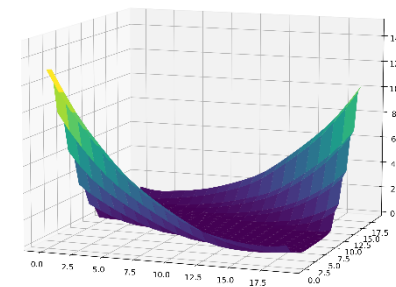
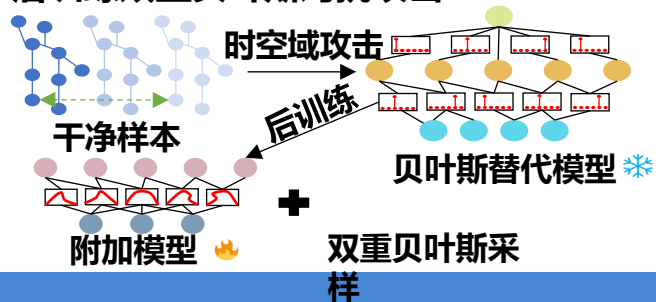
- 时序一致性 (X)
- 损失景观图平滑性 (X)
- 无需重训练 (✓)
- 模型参数尺寸 (X)

贝叶斯对抗攻击



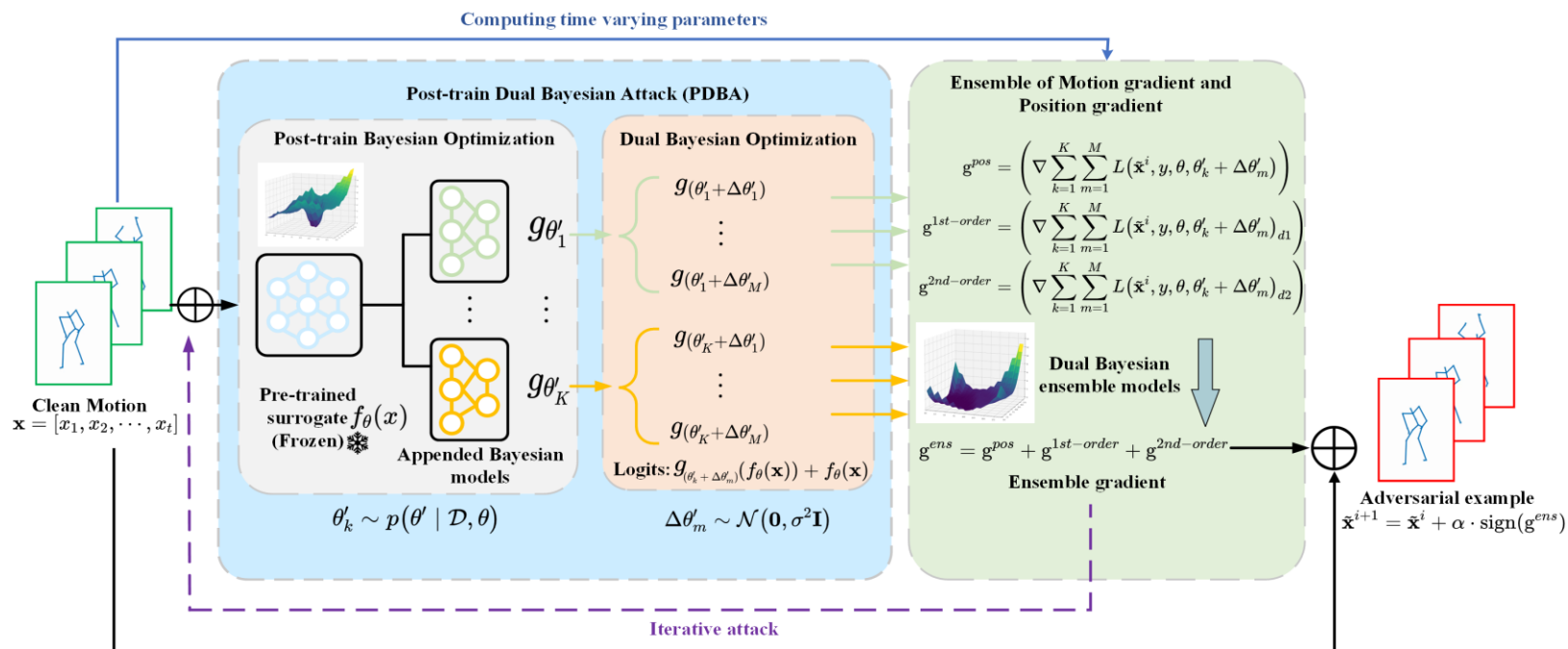
- 时序一致性 (X)
- 损失景观图平滑性 (✓)
- 无需重训练 (X)
- 模型参数尺寸 (X)

后训练双重贝叶斯对抗攻击



- 时序一致性 (✓)
- 损失景观图平滑性 (✓)
- 无需重训练 (✓)
- 模型参数尺寸 (✓)

研究内容



- 将附加模型看作贝叶斯组件并使用一个跳跃连接层将其与预训练模型进行连接。
- 减少采样的时间开销，同时避免了模型重训练。
- 实现对参后验分布的双重采样，增加泛化性，同时结合骨骼的时序特征。

研究结果

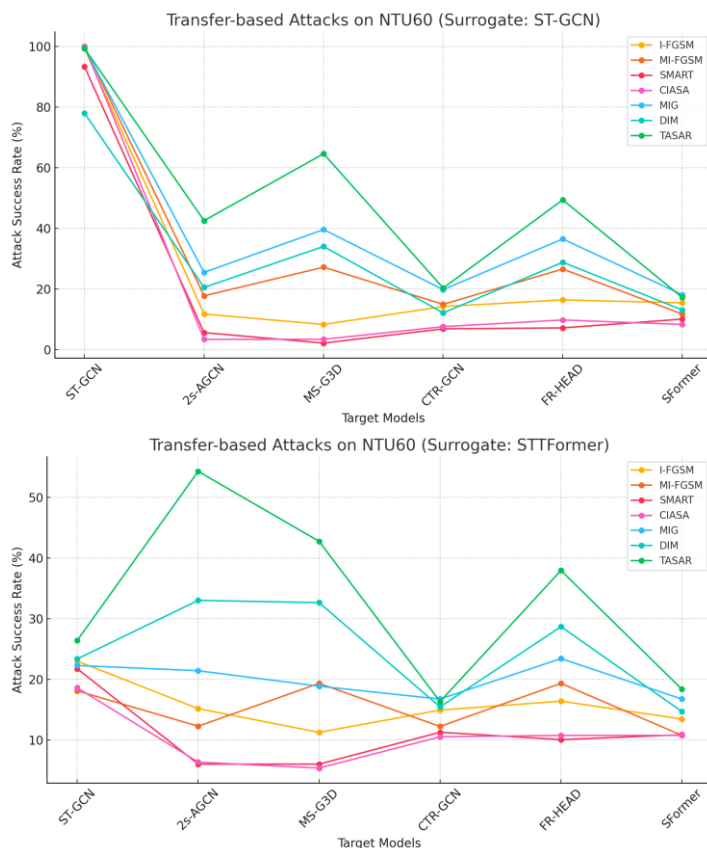


表 有目标攻击迁移攻击成功率，替代模型分别为ST-GCN，MS-G3D

Attack	ST-GCN	2s-AGCN	MS-G3D	CTR-GCN	FR-HEAD
MIFGSM	27.45	3.06	2.32	1.71	1.71
SMART	28.02	1.20	1.81	1.41	1.81
TASAR	28.79	6.06	6.06	8.33	6.82
MIFGSM	2.08	3.31	32.72	1.83	2.45
SMART	0.80	0.60	44.95	1.01	1.01
TASAR	9.09	9.09	57.58	9.85	9.33

➤ 在不同替代模型下，针对同类模型迁移攻击平均成功率最高。

➤ 在正常数据上能够实现杰出性能的检测器仍然面临严重的对抗性威胁，并且改进后攻击方法对这些SOTA检测器具备最强攻击效果。

- 从模型损失景观的角度解释了针对**3D**行为识别的对抗样本迁移性差的原因
- 提出了第一个针对**3D**行为识别的迁移对抗攻击方法**TASAR**
- 建立了第一个骨架行为识别的鲁棒性测评**Benchmark**

目录

CONTENTS

02

后训练黑盒对抗防御方法

基于贝叶斯能量对抗后训练的 黑盒对抗防御方法

刁云峰¹, 姜凯超¹, 郭丹^{1,2*}, 梁振宇^{3,4*}, 时增林¹, 钱振兴⁵, 汪萌^{1,2}

- 1. 合肥工业大学计算机与信息学院, 合肥 230009
 - 2. 合肥综合性国家科学中心人工智能研究院, 合肥 230094
 - 3. 国防科技大学电子对抗学院, 合肥 230038
 - 4. 合肥综合性国家科学中心信息安全研究中心, 合肥 230031
 - 5. 复旦大学计算机科学技术学院, 上海 200433
- * 通信作者. E-mail: guodan@hfut.edu.cn, liangzy21@mudt.edu.cn

国家自然科学基金 (批准号: 62302139, 62272144, 72188101, 62020106007)、安徽省科技重大专项 (批准号: 202203a05020011)、安徽省
杰出青年科学基金项目 (批准号: 2408085J040)、中央高校基本科研业务费专项资金 (批准号: JZ2023HGTA0202, JZ2023HGQA0101)
国家级大学生创新训练项目 (批准号: 202410359032)

刁云峰, 姜凯超, 郭丹, 梁振宇, 时增林, 钱振兴, 汪萌. 中国科学: 信息科学 3

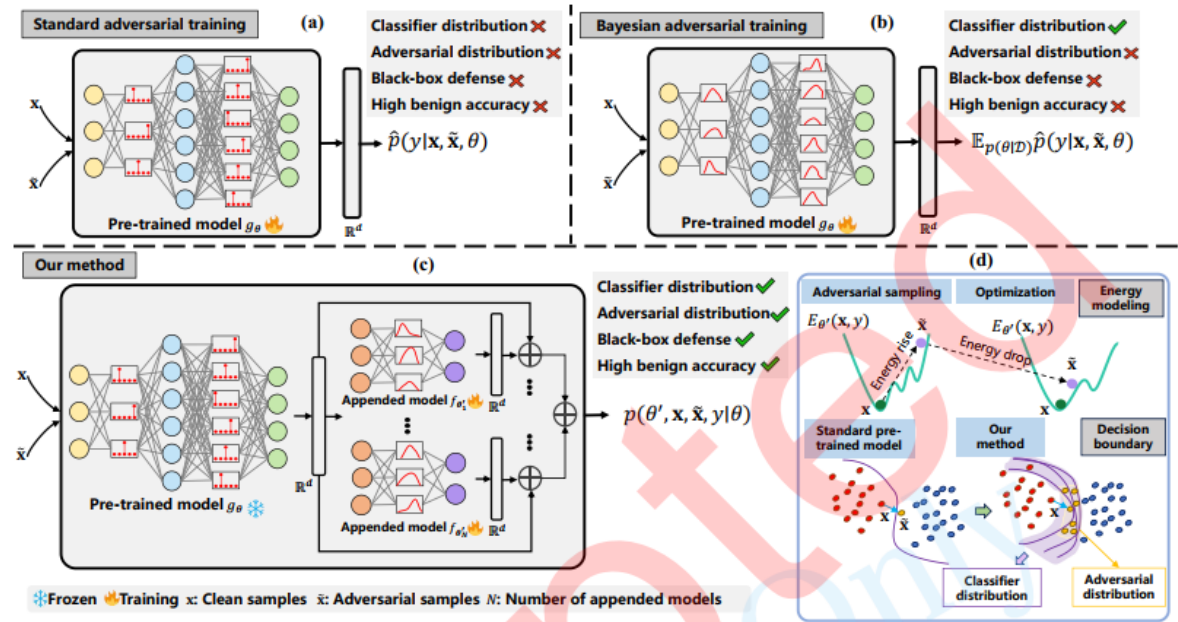


图 1 (网络版彩图) 贝叶斯能量对抗后训练的框架示意图

(Color online) Structure of the Post-train Energy-based Bayesian Adversarial Training

BEAT: Bayesian Energy-based Adversarial Training

Post-train Black-box Defense 通用的黑盒对抗防御方法

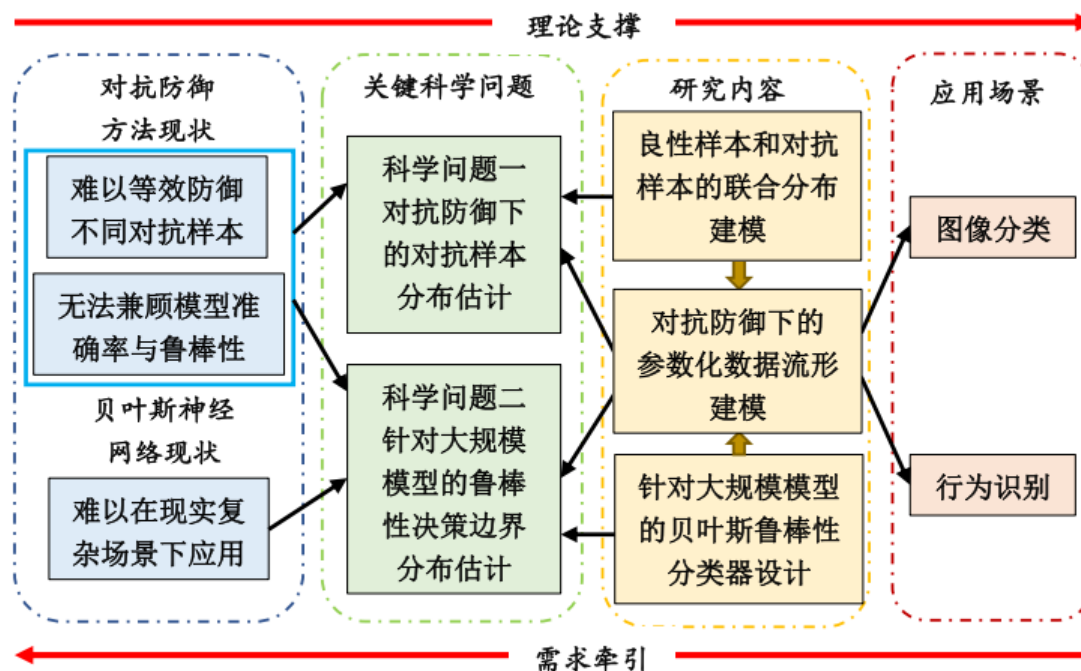
- 第一个后训练黑盒对抗防御方法，不需要重新训练即可让分类器具备鲁棒性
- 通用的贝叶斯能量对抗训练框架，可适用于不同的数据类型（图像分类、行为识别）
- 消除了准确率与鲁棒性之间的权衡问题

[1] He Wang[#], **Yunfeng Diao[#]**, Zichang Tan and Guodong Guo, Defending Black-box Skeleton-based Human Activity Classifiers, AAAI 2023. (CCF-A)

[2] He Wang[#], **Yunfeng Diao[#]**, Post-train Black-box Defense via Bayesian Boundary Correction, TPAMI (under review)

传统对抗防御方法：

- ❑ 白盒对抗防御，模型需要重头进行鲁棒性训练
- ❑ 通常只适用于特定的数据类型
- ❑ 点估计 VS. 分布估计



研究动机

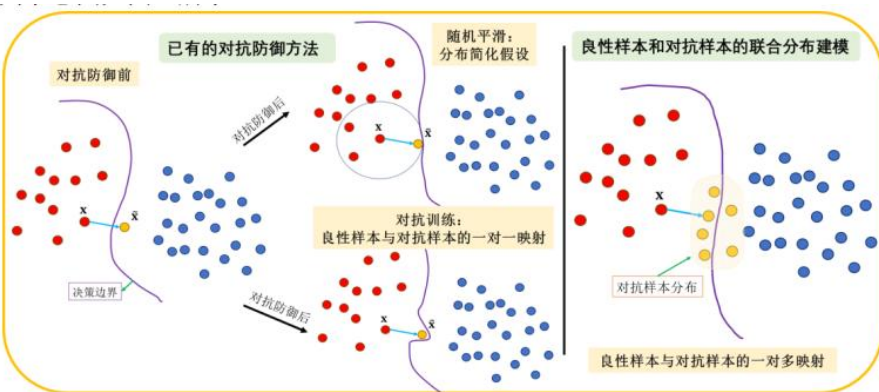


图 5 良性样本和对抗样本的联合分布建模示意图

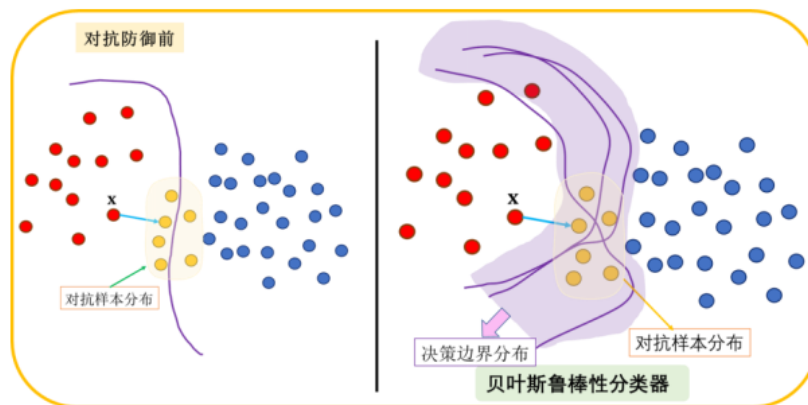


图 6 贝叶斯鲁棒性分类器示意图

实现对数据流型、对抗样本分布、模型参数分布的**全贝叶斯**对待！

Method:

2. Joint Bayesian Modeling of Data and Adversaries

Assuming there is an adversarial distribution over **all** possible attackers, and it can be **observed** during training!



hard to depict directly



all adversarial samples are **close** to the clean data



explain from energy perspective

Adversarial samples should also have relatively **low energy** like clean samples



extend $p_{\theta}(\mathbf{x}, y)$ to $p_{\theta}(\mathbf{x}, \tilde{\mathbf{x}}, y)$ from energy perspective

$$p_{\theta}(\mathbf{x}, \tilde{\mathbf{x}}, y) = \frac{\exp(-E_{\theta}(\mathbf{x}))}{Z(\theta)} = \frac{\exp\{g_{\theta}(\mathbf{x})[y] + g_{\theta}(\tilde{\mathbf{x}})[y] - \lambda d(\mathbf{x}, \tilde{\mathbf{x}})\}}{Z(\theta)}$$

Method:

2. Joint Bayesian Modeling of Data and Adversaries

$$p_{\theta}(\mathbf{x}, \tilde{\mathbf{x}}, y) = p_{\theta}(\tilde{\mathbf{x}} | \mathbf{x}, y) p_{\theta}(\mathbf{x}, y)$$

$$p_{\theta}(\tilde{\mathbf{x}} | \mathbf{x}, y) = \frac{p_{\theta}(\mathbf{x}, \tilde{\mathbf{x}}, y)}{p_{\theta}(\mathbf{x}, y)} = \exp\{g_{\theta}(\tilde{\mathbf{x}})[y] - \lambda d(\mathbf{x}, \tilde{\mathbf{x}})\}$$



a full Bayesian treatment of all adversarial samples!

Method:

3. Bayesian Perspective on the Classifier

$p_{\theta}(\mathbf{x}, \tilde{\mathbf{x}}, y)$: a *point estimation* with respect to θ



$$p(\theta, \mathbf{x}, \tilde{\mathbf{x}}, y) = p(\mathbf{x}, \tilde{\mathbf{x}}, y | \theta) p(\theta)$$



Bayesian Model Averaging

$$p(y' | \mathbf{x}', \mathbf{x}, \tilde{\mathbf{x}}, y) = E_{\theta \sim p(\theta)} [p(y' | \mathbf{x}', \mathbf{x}, \tilde{\mathbf{x}}, y, \theta)]$$

$$\approx \frac{1}{N} \sum_{i=1}^N p(y' | \mathbf{x}', \theta_i), \theta \sim p(\theta | \mathbf{x}, \tilde{\mathbf{x}}, y)$$

huge computational overhead!



post-train Bayesian treatment

$$p(y' | \mathbf{x}', \mathbf{x}, \tilde{\mathbf{x}}, y) = E_{\theta' \sim p(\theta')} [p(y' | \mathbf{x}', \mathbf{x}, \tilde{\mathbf{x}}, y, \theta, \theta')]$$

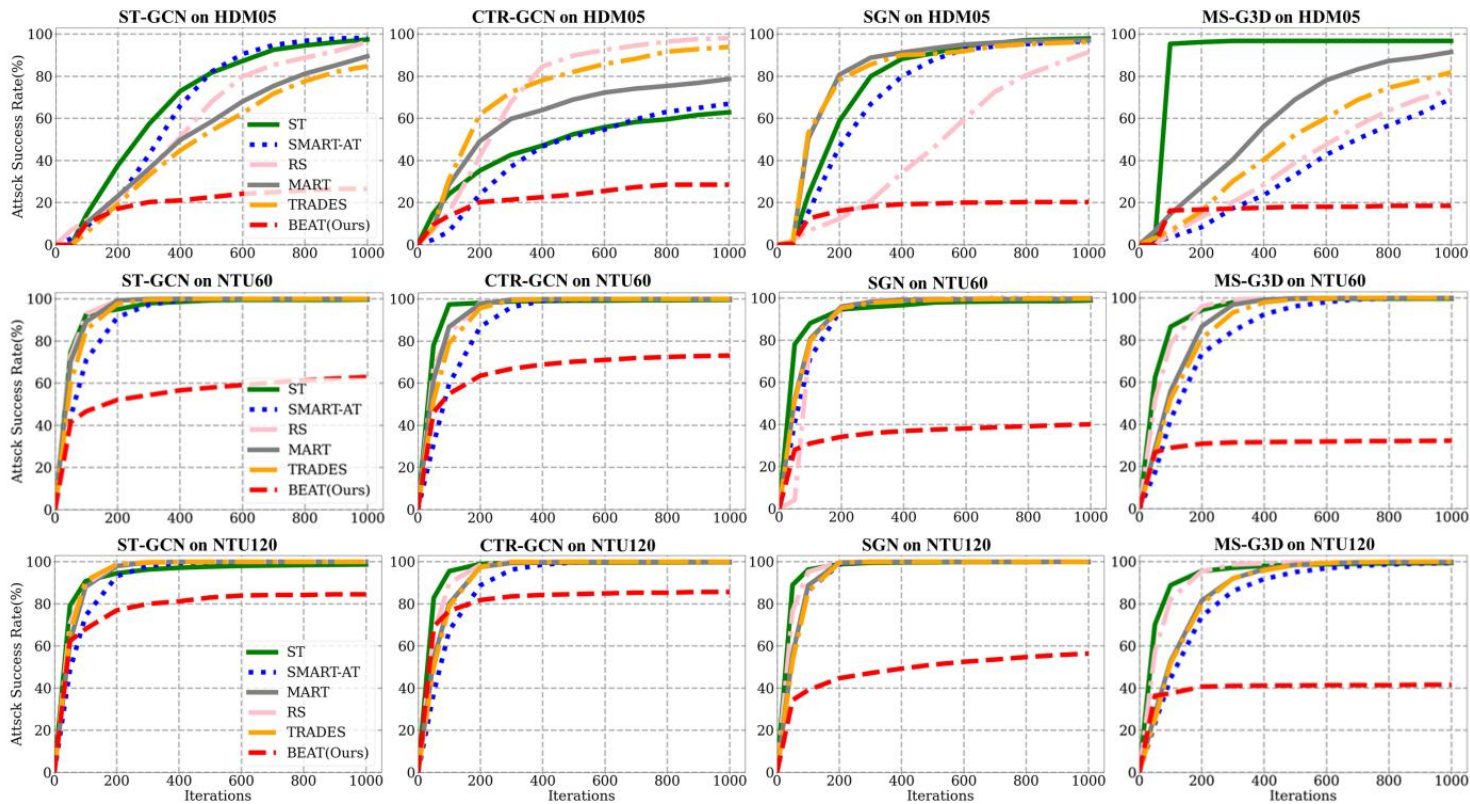
$$\approx \frac{1}{N} \sum_{i=1}^N p(y' | \mathbf{x}', \theta'_i), \theta' \sim p(\theta' | \mathbf{x}, \tilde{\mathbf{x}}, y, \theta)$$

keep the pre-trained classifier *intact* and append a tiny model with parameters θ' behind the classifier

实验结果:

1. 在行为识别任务上的鲁棒性

The attack success rate vs. attack strength curves against SMART



BEAT can defend against more aggressive attacks!

实验结果

1. 在行为识别任务上的鲁棒性

不同防御方法的训练时间对比

Table 6 Training time(hours) of all defense methods

Models	AT	RS	MART	TRADES	BBC
HDM05					
ST-GCN	10.1	1	7.7	10.2	1.2
CTR-GCN	10.1	1.1	10.2	15.4	1.2
SGN	1.4	0.1	1.4	1.1	0.7
MS-G3D	22	1.5	14	16.6	3.4
NTU 60					
ST-GCN	51.7	5.8	49	66.4	34.2
CTR-GCN	93.4	5.6	85.5	96	73.7
SGN	16.7	2.1	11.9	13.1	10.1
MS-G3D	240	23.9	132	217	72
NTU 120					
ST-GCN	29.4	2.1	28.2	35.8	19.7
CTR-GCN	51.6	2.6	52.5	51.7	23.4
SGN	27.5	5.6	18.2	20.5	4.6
MS-G3D	378.3	46.1	214	322	115.3

实验结果:

1. 在图像分类任务上的鲁棒性

TABLE 2

Comparing robustness(%) under different EoT-PGD attack budget.

Model&Data	Defenses	0	0.035	0.055	0.07
VGG-16 on CIFAR-10	None	93.6	0	0	0
	PGD-AT	80.3	31.1	15.5	10.3
	Adv-BNN	79.7	37.7	16.3	8.1
	IG-BNN	83.6	50.2	26.8	16.9
	BBC(Ours)	93.3	79.1	63.1	48.4
ModelA on STL-10	None	78.5	0	0	0
	PGD-AT	63.2	27.4	12.8	7.0
	Adv-BNN	59.9	31.4	16.7	9.1
	IG-BNN	64.3	48.2	34.9	27.3
	BBC(Ours)	78.2	43.3	41.2	39.3

Comparing robustness(%) on CIFAR-10 and CIFAR-100.

Method	CIFAR-10			CIFAR-100		
	Clean	PGD-20	PGD-50	Clean	PGD-20	PGD-50
ST	96.2	0	0	80.4	0	0
PGD-AT	85.2	55.1	54.9	60.9	31.7	31.5
TRADES	85.7	56.1	55.9	58.6	28.7	26.6
MART	84.2	58.6	58.1	60.8	26.4	25.8
LAS-AT	87.7	60.2	59.8	64.9	36.4	36.1
AWP	85.6	58.1	57.9	60.4	33.9	33.7
FAT	88.0	49.9	48.8	-	-	-
LBGAT	88.2	54.7	54.3	60.6	34.8	34.6
RAT	86.1	61.4	-	64.7	35.7	-
BBC(Ours)	94.7	93.9	93.7	74.4	70.4	69.5